

Asymptotic Preserving Semi-Lagrangian Discontinuous Galerkin Methods for Multiscale Kinetic Transport Equations

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Abstract

We propose a new family of asymptotic preserving (AP) discontinuous Galerkin (DG) schemes for kinetic transport equations under a diffusive scaling. They are built upon a characteristics-based model reformulation proposed in [Zhang, Zhu and Xiong, SIAM J. Sci. Comput., 2023]. The reformulated model comprises a convection-diffusion-like equation for the macroscopic density and another evolution equation for the distribution function. We employ the equations in a weak form to enjoy the advantages of a DG method. We also leverage a semi-Lagrangian (SL) approach to achieve higher efficiency than a fully implicit scheme without loss of unconditional stability. A damping technique is introduced to control spurious numerical oscillations for high-order DG methods. We establish formal AP and Fourier stability analyses for fully discrete schemes in a linear scenario. Numerical experiments, covering one-dimensional to three-dimensional in space problems, confirm the accuracy, AP property, uniformly unconditional stability, and high parallel efficiency of the proposed methods.

Keywords: kinetic transport equation, diffusive scaling, asymptotic preserving, semi-Lagrangian, discontinuous Galerkin, parallel computing

1. Introduction

Kinetic theory is very important in multi-scale modeling, which provides a fundamental framework to bridge the gap between microscopic and macroscopic scales. It has been widely used in neutron transport, radiative transport, rarefied gas dynamics, and other related fields [1, 2]. In this work, we are interested in the kinetic transport equation under a diffusive scaling

$$\varepsilon \partial_t f + \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{1}{\varepsilon} \mathcal{Q}(f). \quad (1.1)$$

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7 Here the unknown $f(t, \mathbf{x}, \mathbf{v})$, depending on time $t \in \Omega_t \subset \mathbb{R}^+$, spatial position $\mathbf{x} \in \Omega_{\mathbf{x}} \subset \mathbb{R}^d$
8 and velocity $\mathbf{v} \in \Omega_{\mathbf{v}} \subset \mathbb{R}^d$ where $1 \leq d \leq 3$, represents the density distribution function of
9 particles in the phase space. The operator $\mathcal{Q}$ describes the collision effect among particles or
10 between particles and the medium. $\varepsilon > 0$ is the Knudsen number, defined as the ratio of a
11 mean free path to a typical spatial length, determining the flow regime. With the presence of ε
12 in front of the time derivative of f , we focus on the long-time behavior of the physical system.

13 Developing an efficient numerical scheme for this high-dimensional kinetic problem (espe-
14 cially when $d = 3$) entails several challenges. Firstly, we need to design a scheme that is effective
15 not only in the kinetic regime (when ε is away from zero) but also in the diffusive regime (when ε
16 is close to zero). A naive numerical discretization might produce unphysical solutions in the dif-
17 fusive limit [3, 4, 5]. One popular successful approach is to design asymptotic-preserving (AP)
18 schemes [6, 7, 8]. The basic idea behind AP schemes is to ensure that the numerical method
19 accurately captures the asymptotic limit of the mathematical model at the discrete level. For
20 (1.1), an AP scheme must be a consistent discretization for the corresponding limiting diffusion
21 equation as $\varepsilon \rightarrow 0$. There are various ways to construct an AP scheme, including diffusive
22 relaxation method [9, 10], penalization method [11], macro-micro decomposition [12, 13], and
23 exponential integration [14, 15, 16]. Secondly, both the transport and collision terms exhibit
24 strong stiffness when ε is small, posing great challenges to efficient time discretization, espe-
25 cially in the high-dimensional case. On the one hand, a fully explicit scheme yields a stringent
26 time step condition $\Delta t = \mathcal{O}(\varepsilon h)$ (h is the mesh size) for stability and accuracy [17, 18]. On
27 the other hand, although a fully implicit scheme can avoid a stability limitation on the time
28 step, it is very costly to solve a large-scale system of linear equations at each time step, even
29 with efficient iterative solvers [19, 20]. In addition, several implicit-explicit (IMEX) schemes
30 are developed to balance stability conditions and computational cost. For stability, AP IMEX
31 schemes (e.g., [12, 13]) require a parabolic time step $\Delta t = \mathcal{O}(h^2)$ when $\varepsilon \ll h$ and $\Delta t = \mathcal{O}(\varepsilon h)$
32 otherwise, see [21, 22] for some rigorous analyses. To enhance stability in the diffusive regime,
33 one can apply an explicit-implicit-null technique [10, 23, 24] or modify an existing AP scheme
34 by implicitly discretizing diffusion-related terms [15, 16, 25, 26]. However, it is worth noting
35 that when the system departs from the diffusive regime, these stability-enhanced AP schemes
36 retain ε -dependent stability condition $\Delta t = \mathcal{O}(\frac{\varepsilon^2}{\varepsilon/h - \lambda})$. When the system enters the intermediate
37 regime with $\varepsilon = \mathcal{O}(h)$, the time step condition becomes $\Delta t = \mathcal{O}(h^2)$ again. This restriction is
38 still not pleasant in real applications, particularly when the regime varies in the computational
39 domain, for example, problems with a large-range-variant opacity [24, 27].

40 To overcome the stability issue in the intermediate regime, Zhang et al [27] propose a new
41 approach to reformulate the model (1.1), and then design a class of finite difference schemes
42 that is uniformly unconditionally stable in all regimes. In such a framework, an approxima-
43 tion model for the macroscopic density is derived by following characteristics and recovering
44 both convection and diffusion processes. For the numerical scheme, the density is updated
45 using a semi-Lagrangian (SL) finite difference scheme for convection and a linearly implicit
46 finite difference discretization for diffusion. The resulting scheme is verified to be uniformly
47 unconditionally stable based on a Fourier analysis. Once the macroscopic density is available,
48 the high-dimensional distribution function is fully decoupled in velocity via a discrete veloc-
49 ity method¹ [28, 29]. This decoupled updating is very convenient for parallel computing to
50 avoid data racing. Notably, the kinetic equation is solved at each discrete velocity via a lin-

¹In the radiative transfer community, this is often referred to as the discrete ordinate method or S_N method.

early implicit finite difference scheme, which involves a set of small-scale sparse linear systems (only depends on the spatial discretization). This approach is also used to design an efficient Monte-Carlo method for radiative transfer equations [30].

In this work, we aim to develop an AP SL discontinuous Galerkin (DG) method for kinetic transport problems (1.1). Here in space, we use a discontinuous finite element approximation in a weak form. The original DG scheme was developed by Reed and Hill for neutron transport equations [31]. Since then, due to its advantages of compactness, easy h - p adaptivity, and flexibility in handling complicated geometries, it has been widely used to study stationary and non-stationary problems, e.g., [32, 33, 34, 35, 36, 37, 38]. In time, we follow characteristics by fully utilizing an SL framework. Besides an SL flux approximation for the density equation, we also employ an SL approach to solve the kinetic equation (1.1), inspired by [39, 40, 41, 42]. For the diffusion part, a linearly implicit local DG method is used. In this case, we only need to solve one small-scale sparse linear system at each step or each stage. Compared to the finite difference scheme in [27], the discontinuous finite element method in a weak form has several advantages. Firstly, it is flexible for h - p adaptivity and able to be applied to irregular geometries with unstructured meshes. Secondly, in order to control the numerical oscillation for the second order DG method, we introduce a damping term as proposed in [43, 44] instead of a slope limiter used in [27]. The damping approach is more convenient for implicit time discretization and performs better than a slope limiter in the diffusive regime. Thirdly, in the weak form, it is possible to perform an energy stability analysis as has been done in [22, 23, 45]. The new contribution of this work is how to handle the SL characteristic tracing in a finite element weak form for both the transport part in the density equation and the kinetic equation. The resulting scheme is more efficient and flexible than the previous finite difference scheme. At this moment, we still only establish formal AP and Fourier stability analyses for a fully discrete scheme of the two-velocity telegraph equation. An energy stability analysis is left as our ongoing work. Numerical experiments, from one-dimensional (1D) to three-dimensional (3D) in space problems, are performed to validate the accuracy, stability, and high parallel efficiency of our proposed approach.

The rest of the paper is organized as follows. In Section 2, we revisit the model equations and derive the approximation model in the general multi-dimensional case. The numerical methods for the approximation model are described in Section 3. Formal asymptotic and Fourier stability analyses are given in Section 4. We provide some numerical tests in Section 5, followed by a conclusion remark in Section 6.

2. Model reformulation

Let $\Omega_{\mathbf{v}} \subset \mathbb{R}^d$ be bounded and symmetric with respect to the origin of $\mathbb{R}^d$. Suppose that $f(\mathbf{v})$ locates in the Hilbert space $L^2(\Omega_{\mathbf{v}}; d\mu)$, where μ is a Lebesgue measure such that $\int_{\Omega_{\mathbf{v}}} 1 d\mu = 1$. We denote $\langle f_1, f_2 \rangle := \int_{\Omega_{\mathbf{v}}} f_1 f_2 d\mu$ as the $L^2(\Omega_{\mathbf{v}})$ inner product, and $\langle f \rangle := \langle f, 1 \rangle$ as the average in $\Omega_{\mathbf{v}}$ of any μ -measurable function f . Due to the symmetry of $\Omega_{\mathbf{v}}$, we have that $\langle g \rangle = 0$ for any odd function $g(\mathbf{v})$.

In this paper, we are interested in two classes of kinetic transport equations. The first class is the two-velocity model in the 1D case. Let $\Omega_{\mathbf{v}} = \{-1, 1\}$ and μ be a discrete measure. The

92 following four collision operators are widely considered [6, 13]:

$$\mathcal{Q}(f) = \langle f \rangle - f; \quad (2.1a)$$

$$\mathcal{Q}(f) = \langle f \rangle - f + A\varepsilon v \langle f \rangle, \quad |A\varepsilon| < 1; \quad (2.1b)$$

$$\mathcal{Q}(f) = \langle f \rangle - f + B\varepsilon v [\langle f \rangle^2 - (\langle f \rangle - f)^2], \quad B > 0; \quad (2.1c)$$

$$\mathcal{Q}(f) = K \langle f \rangle^m (\langle f \rangle - f), \quad K > 0, \quad m \leq 0. \quad (2.1d)$$

93 In particular, the kinetic equation (1.1) with collision (2.1a) is named the Goldstein-Taylor
 94 model or the telegraph equation. The second class consists of multi-dimensional linear transport
 95 equations with a collision operator

$$\mathcal{Q}(f) = \sigma_s(\langle f \rangle - f) - \varepsilon^2 \sigma_a f + \varepsilon^2 S, \quad (2.2)$$

96 where $\sigma_s(\mathbf{x}) > 0, \sigma_a(\mathbf{x}) \geq 0$, and $\sigma = \sigma_s + \varepsilon^2 \sigma_a$ are associated with scattering, absorption,
 97 and total transport coefficients, respectively. The term $S(\mathbf{x})$ represents the internal source of
 98 particles. In this case, the average operator can be written as $\langle \cdot \rangle = \frac{1}{|\Omega_v|} \int_{\Omega_v} \cdot d\mathbf{v}$, where

$$\begin{aligned} \Omega_v &= [-1, 1], & |\Omega_v| &= 2, & \text{for } d &= 1; \\ \Omega_v &= \{\mathbf{v} : v_x^2 + v_y^2 = 1\}, & |\Omega_v| &= 2\pi, & \text{for } d &= 2; \\ \Omega_v &= \{\mathbf{v} : v_x^2 + v_y^2 + v_z^2 = 1\}, & |\Omega_v| &= 4\pi, & \text{for } d &= 3. \end{aligned} \quad (2.3)$$

99 Such equations arise in neutron transport [46], radiative transfer [1], wave transport in random
 100 media [47], etc.

101 2.1. Macro-Micro reformulation

102 The macro-micro decomposition provides an elegant way to reformulate the linear kinetic
 103 equation (1.1) and analyze their asymptotic properties [12]. The basic idea is to use an orthog-
 104 onal decomposition of f :

$$f = \Pi f + (I - \Pi)f = \rho + \varepsilon g, \quad \rho := \langle f \rangle, \quad \langle g \rangle = 0, \quad (2.4)$$

105 where I is the identity operator and $\Pi(\cdot) = \langle \cdot \rangle$ is an orthogonal operator from $L^2(\Omega_v; d\mu)$
 106 onto the subspace spanned by all functions independent of $\mathbf{v}$. By acting Π and its orthogonal
 107 complement $I - \Pi$ on (1.1), we obtain

$$\partial_t \rho + \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} g \rangle = \frac{1}{\varepsilon^2} \Pi(\mathcal{Q}(\rho + \varepsilon g)), \quad (2.5a)$$

$$\partial_t g + \frac{1}{\varepsilon} (I - \Pi)(\mathbf{v} \cdot \nabla_{\mathbf{x}} g) + \frac{1}{\varepsilon^2} \mathbf{v} \cdot \nabla_{\mathbf{x}} \rho = \frac{1}{\varepsilon^3} (I - \Pi)(\mathcal{Q}(\rho + \varepsilon g)). \quad (2.5b)$$

108 The solvability of f in (1.1) is equivalent to that of ρ and g in (2.5). Specifically, one can verify
 109 that as $\varepsilon \rightarrow 0$, the limiting density equation for collision operators (2.1a)-(2.1d) are

$$\partial_t \rho = \langle v^2 \rangle \partial_{xx} \rho, \quad \text{heat equation;} \quad (2.6a)$$

$$\partial_t \rho + A \langle v^2 \rangle \partial_x \rho = \langle v^2 \rangle \partial_{xx} \rho, \quad \text{advection-diffusion equation;} \quad (2.6b)$$

$$\partial_t \rho + B \langle v^2 \rangle \partial_x (\rho^2) = \langle v^2 \rangle \partial_{xx} \rho, \quad \text{viscous Burgers equation;} \quad (2.6c)$$

$$\partial_t \rho = \frac{\langle v^2 \rangle}{K(1-m)} \partial_{xx} \rho, \quad \text{porous media equation.} \quad (2.6d)$$

110 For collision operator (2.2), the density ρ also obeys a diffusion equation when $\varepsilon \rightarrow 0$,

$$\partial_t \rho - \nabla_{\mathbf{x}} \cdot (\sigma_s^{-1} \mathbf{D} \nabla_{\mathbf{x}} \rho) = -\sigma_a \rho + S, \quad \mathbf{D} = \langle \mathbf{v} \otimes \mathbf{v} \rangle. \quad (2.7)$$

111 Here $\sigma_s^{-1} \mathbf{D}$ denotes a $d \times d$ diffusion coefficient matrix, and $\otimes$ is a Kronecker product. Partic-
 112 ularly, with (2.3), the matrix $\mathbf{D}$ becomes a diagonal matrix $D\mathbf{I}$ where D is a constant and $\mathbf{I}$ is
 113 a $d \times d$ identity matrix. Specifically, we deduce $D = \frac{1}{3}$ for $d = 1, 3$ and $D = \frac{1}{2}$ for $d = 2$.

114 With appropriate IMEX time discretization strategies [12, 13], one can solve the macro-
 115 micro system (2.5) in a sequentially explicit way which leads to an AP scheme. Furthermore,
 116 unconditional stability in the diffusive regime can be achieved by utilizing an extra artificial
 117 viscosity or based on Schur decomposition [23, 26]. However, all the mentioned schemes treat the
 118 stiff nonlinear convection term $\frac{1}{\varepsilon}(I - \Pi)(\mathbf{v} \cdot \nabla_{\mathbf{x}} g)$ explicitly. This causes a time step restriction in
 119 the intermediate regime when this convection term is comparable to the diffusion term [23, 45].
 120 If we apply an implicit time discretization to this convection term, (2.5) would give rise to a
 121 large-scale linear system that couples ρ and g , thus introducing additional complexity compared
 122 to the original equation (1.1).

123 2.2. Semi-Lagrangian reformulation

124 To ensure a clear and concise description, we focus on the following equation with the
 125 simplest collision operator:

$$\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{1}{\varepsilon^2} (\langle f \rangle - f). \quad (2.8)$$

126 Here $\Omega_{\mathbf{v}}$ is either a two-velocity space or a continuous velocity space in (2.3). To overcome
 127 the limitation of a macro-micro decomposition system, we start with a macro-micro system
 128 comprising the following set of equations for ρ and f :

$$\partial_t \rho + \frac{1}{\varepsilon} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} f \rangle = 0, \quad (2.9a)$$

$$\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{1}{\varepsilon^2} (\rho - f). \quad (2.9b)$$

129 As approaching the equilibrium, the microscopic equation (2.9b) becomes $f = \rho$ for all $\mathbf{v} \in \Omega_{\mathbf{v}}$
 130 and the macroscopic equation (2.9a) cannot converge to the correct asymptotic limit. To
 131 capture the diffusive limit, the key idea is to reformulate the macroscopic flux $\mathbf{F} := \frac{1}{\varepsilon} \langle \mathbf{v} f \rangle$,
 132 which should recover a diffusion flux for the diffusive limit [27, 30].

133 To begin with, we rewrite the kinetic equation (2.9b) in the following characteristic form

$$\frac{df}{dt} = \frac{1}{\varepsilon^2} (\rho - f), \quad (2.10a)$$

$$\frac{d\mathbf{x}}{dt} = \frac{\mathbf{v}}{\varepsilon}, \quad (2.10b)$$

134 with d/dt being a material derivative. Without loss of generality, we set $t \in [t_0, t_1]$. For
 135 notations, we denote $\mathbf{x}_a = \mathbf{x}(t_a)$ for $a = 0, 1$ and $u(t) = u(t, \mathbf{x}(t))$ for $u = f, \rho, \partial_t \rho, \nabla_{\mathbf{x}} \rho$.
 136 Starting from $t = t_1$, a backward tracing of the characteristic line is

$$\mathbf{x}(t) = \mathbf{x}_1 - \frac{t_1 - t}{\varepsilon} \mathbf{v}, \quad \forall t \in [t_0, t_1]. \quad (2.11)$$

137 By multiplying (2.10a) with an exponential factor e^{t/ε^2} and integrating over $[t_0, t_1]$, we obtain

$$f(t_1) = e^{-(t_1-t_0)/\varepsilon^2} f(t_0) + \int_{t_0}^{t_1} \frac{1}{\varepsilon^2} e^{-(t_1-t)/\varepsilon^2} \rho(t) dt. \quad (2.12)$$

138 Integrating by parts for the time integral in (2.12), we get

$$f(t_1) = \rho(t_1) + e^{-(t_1-t_0)/\varepsilon^2} (f(t_0) - \rho(t_0)) - \int_{t_0}^{t_1} e^{-(t_1-t)/\varepsilon^2} \frac{d\rho}{dt}(t) dt. \quad (2.13)$$

139 According to (2.4), we have $g = (f - \rho)/\varepsilon$, and it yields

$$g(t_1) = e^{-(t_1-t_0)/\varepsilon^2} g(t_0) - \frac{1}{\varepsilon} \int_{t_0}^{t_1} e^{-(t_1-t)/\varepsilon^2} \frac{d\rho}{dt}(t) dt. \quad (2.14)$$

140 For the time integral in (2.14), applying the following approximation,

$$\int_{t_0}^{t_1} e^{-(t_1-t)/\varepsilon^2} \frac{d\rho}{dt}(t) dt \approx \left(\int_{t_0}^{t_1} e^{-(t_1-t)/\varepsilon^2} dt \right) \frac{d\rho}{dt} \Big|_{t=t_1} = \varepsilon^2 (1 - e^{-(t_1-t_0)/\varepsilon^2}) \frac{d\rho}{dt} \Big|_{t=t_1}, \quad (2.15)$$

141 together with

$$\frac{d\rho}{dt}(t) = \partial_t \rho(t) + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} \rho(t), \quad (2.16)$$

142 we deduce

$$g(t_1) = e^{-(t_1-t_0)/\varepsilon^2} g(t_0) - \varepsilon (1 - e^{-(t_1-t_0)/\varepsilon^2}) \partial_t \rho(t_1) - (1 - e^{-(t_1-t_0)/\varepsilon^2}) (\mathbf{v} \cdot \nabla_{\mathbf{x}} \rho)(t_1). \quad (2.17)$$

143 By taking the first moment of (2.17) in $\Omega_{\mathbf{v}}$, we obtain the approximation flux $\mathbf{G}$ at time t_1 :

$$\mathbf{F}(t_1) = \frac{1}{\varepsilon} \langle \mathbf{v} f(t_1) \rangle = \langle \mathbf{v} g(t_1) \rangle \approx \mathbf{G}(t_1) := e^{-(t_1-t_0)/\varepsilon^2} \langle \mathbf{v} g(t_0) \rangle - (1 - e^{-(t_1-t_0)/\varepsilon^2}) (\mathbf{D} \nabla_{\mathbf{x}} \rho)(t_1). \quad (2.18)$$

144 Here $\mathbf{D}$ is the same as in (2.7). As we can see, the first term contributes to the convection
 145 process, while the second term is associated with the diffusion process. We can easily verify
 146 that when $\varepsilon \rightarrow 0$, the contribution of convection diminishes, while $\mathbf{G}$ converges to the diffusion
 147 flux, i.e., $\mathbf{G} \rightarrow \mathbf{D} \nabla_{\mathbf{x}} \rho$.

148 Suppose we would like to integrate (2.9) up to a fixed time T . We assume discrete time
 149 steps t^n ($n = 0, 1, \dots$) for the time interval $[0, T]$, and take a uniform step size $\Delta t = t^{n+1} - t^n$.
 150 In the SL framework, particles are tracked within a single time step, rather than throughout
 151 their entire motion in the fully Lagrangian framework. Setting $t_1 = t$, $\mathbf{x}_1 = \mathbf{x}$, $t_0 = t^n$, $\mathbf{x}_0 =$
 152 $\mathbf{x}^n = \mathbf{x} - \mathbf{v}(t - t^n)/\varepsilon$ and substituting (2.18) into (2.9a), we obtain an SL approximation for
 153 (2.9):

$$\partial_t \rho + e^{-(t-t^n)/\varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} g(t^n, \mathbf{x}^n, \mathbf{v}) \rangle = (1 - e^{-(t-t^n)/\varepsilon^2}) \nabla_{\mathbf{x}} \cdot (\mathbf{D} \nabla_{\mathbf{x}} \rho), \quad (2.19a)$$

$$\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{1}{\varepsilon^2} (\rho - f), \quad (2.19b)$$

154 where

$$g(t^n, \mathbf{x}^n, \mathbf{v}) = (f(t^n, \mathbf{x}^n, \mathbf{v}) - \rho(t^n, \mathbf{x}^n))/\varepsilon. \quad (2.20)$$

Our modeling decomposes the transport process into two parts: convection and diffusion. In particular, the convection term involves a backward tracing of microscopic perturbation g along the characteristic line, eliminating a scale-related time step limitation. The weight functions, represented as $\omega_1 = e^{-(t-t^n)/\varepsilon^2}$ and $\omega_2 = 1 - e^{-(t-t^n)/\varepsilon^2}$, balance the contribution of convection and diffusion. When ε is away from zero, both convection and diffusion are taken into account. When ε is close to zero, (2.19) becomes diffusion-dominated and formally converges to

$$\partial_t \rho = \nabla_{\mathbf{x}} \cdot (\mathbf{D} \nabla_{\mathbf{x}} \rho), \quad f = \rho. \quad (2.21)$$

This also shows that the new model preserves the desired diffusive limit. Theorem 2.2 below provides a new modeling error estimate that is more accurate than the one presented in [27]. Furthermore, this new error estimate offers insights into the order reduction observed in our numerical tests, see Section 5. To maintain the m -th order convergence rate, it is necessary to take $\Delta t = \varepsilon^{\frac{3}{2}} h^{\frac{m}{2}}$ and $h^m \leq \varepsilon$, where h is the mesh size. For the sake of readability, we include the proofs for Lemma 2.1 and Theorem 2.2 in the appendix. In addition, extending the proposed modeling approach to other collision operators is feasible but may involve some tricks, see [27] for more details.

Lemma 2.1. *Suppose that a function $\eta = \eta(\tau)$ is smooth enough and there exists a uniform bound $M > 0$ such that $|\beta^{k+1} \eta^{(k)}(\tau)| \leq M$ for all $k \geq 1$ and $\tau \in [0, s]$, where $0 < \beta \leq 1$ is a given constant and $\eta^{(k)}(s)$ is the k -th derivative at a point $s > 0$. For $\alpha > 0$, the following error estimate holds:*

$$\left| \int_0^s e^{-\alpha(s-\tau)} (\eta(\tau) - \eta(s)) d\tau \right| \leq MU(\alpha, \beta, s), \quad (2.22)$$

where

$$U(\alpha, \beta, s) = \begin{cases} e^{-\alpha s} - 1 + \alpha s, & \alpha\beta = 1; \\ \frac{(\alpha\beta)^{-2}(1-e^{-\alpha s})}{1-(\alpha\beta)^{-1}} + \frac{(\alpha\beta)^{-1}(e^{-\alpha s} - e^{-(\alpha-\beta^{-1})s})}{1-(\alpha\beta)^{-1}}, & \alpha\beta \neq 1. \end{cases} \quad (2.23)$$

Theorem 2.2. *Let $0 < \varepsilon \leq 1$, $t \in [t_0, t_1]$ and $\Delta t = t_1 - t_0$. Suppose that there exists a uniform upper bound $M > 0$ such that for all $k \geq 1$ and $t \in [t_0, t_1]$,*

$$|(\varepsilon \partial_t + \mathbf{v} \cdot \nabla_{\mathbf{x}})^{k+1} \rho(t)| \leq M. \quad (2.24)$$

Then, the following error estimate holds for the flux approximation (2.18):

$$\|\mathbf{F} - \mathbf{G}\|_{\infty} \leq CW^*(\varepsilon, \Delta t), \quad W^*(\varepsilon, \Delta t) = \begin{cases} \frac{\Delta t^2}{2\varepsilon^3}, & \text{if } \Delta t \leq \varepsilon^2; \\ \frac{\varepsilon}{1-\varepsilon}, & \text{if } \Delta t > \varepsilon^2, \end{cases} \quad (2.25)$$

where $C > 0$ is a constant independent of ε and Δt but depends on M and velocity space $\Omega_{\mathbf{v}}$.

3. Numerical method

In this section, we will present our methods for the approximation model (2.19). For the convection part in both the macroscopic equation (2.19a) and the microscopic equation (2.19b), we use an SL approach with an upwind DG approximation in space. For the diffusion part in the macroscopic equation (2.19a), an implicit LDG approximation is taken. The main challenging part is the convection in (2.19a): although an implicit time discretization is used, the solution for a DG approximation is defined at the previous time level t^n due to an SL tracking.

185 We first review some notations for a DG method. We assume a quasi-uniform mesh $\mathcal{T}_h$ with
 186 its boundary Γ_h , namely, there exists a constant $c_0 > 0$ such that

$$h \leq c_0 r, \quad h = \max_{K \in \mathcal{T}_h} h_K, \quad r = \max_{K \in \mathcal{T}_h} r_K, \quad (3.1a)$$

$$h_K = \text{diam}(K), \quad r_K = \max \{ \text{diam}(S); S \text{ is a ball contained in } K \}. \quad (3.1b)$$

187 Then we consider a piecewise polynomial space on $\mathcal{T}_h$ as

$$U_h^k = \{u \in L^2(\Omega_{\mathbf{x}}); u|_K \in \mathcal{P}^k(K), \quad \forall K \in \mathcal{T}_h\}. \quad (3.2)$$

188 Here $\mathcal{P}^k(K)$ is the set of all polynomials of degree at most k on K . We also introduce

$$V_h^k = \left\{ u(\cdot, \mathbf{v}) \in U_h^k; \int_{\Omega_{\mathbf{v}}} \int_{\Omega_{\mathbf{x}}} |u(\mathbf{x}, \mathbf{v})|^2 d\mathbf{x} d\mathbf{v} < \infty \right\}. \quad (3.3)$$

189 For a point $\mathbf{x}$ on the cell boundary ∂K with a unit outer normal vector $\mathbf{n}$, we denote its
 190 pointwise limits along $\mathbf{n}$ and $-\mathbf{n}$ by $\mathbf{x}^+$ and $\mathbf{x}^-$, respectively. We also define the jump and
 191 average of a function $u \in U_h^k$ at $\mathbf{x}$ along the normal vector as

$$[[u]](\mathbf{x}) = u(\mathbf{x}^+) - u(\mathbf{x}^-), \quad \{u\}(\mathbf{x}) = \frac{1}{2} (u(\mathbf{x}^+) + u(\mathbf{x}^-)). \quad (3.4)$$

192 *3.1. An implicit LDG method for the macroscopic equation*

193 The macroscopic equation (2.19a) is close to a convection-diffusion equation, where the
 194 convection term has already been traced back to time level t^n by an SL method. First, we
 195 rewrite (2.19a) as a system only containing first-order derivatives

$$\mathbf{q} = \mathbf{D} \nabla_{\mathbf{x}} \rho, \quad (3.5a)$$

$$\partial_t \rho + e^{-(t-t^n)/\varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} \mathcal{S}_{\mathbf{v}}^n[g] \rangle = (1 - e^{-(t-t^n)/\varepsilon^2}) \nabla_{\mathbf{x}} \cdot \mathbf{q}. \quad (3.5b)$$

196 Here for clarity of notation, we introduce an SL evolution operator

$$\mathcal{S}_{\mathbf{v}}^n[g](\mathbf{x}, \mathbf{v}) = g(t^n, \mathbf{x} - \mathbf{v} \Delta t / \varepsilon, \mathbf{v}). \quad (3.6)$$

197 With superscript n , we emphasize that $\mathcal{S}_{\mathbf{v}}^n[g]$ is determined by the values of g at time level t^n .

198 We start with a temporal discretization. In order to overcome the parabolic time stability
 199 condition in the diffusive regime, we intend to treat the diffusion term implicitly. Besides, the
 200 convection term will be treated via a backward SL approach to eliminate stiffness. For high-
 201 order time discretizations, we utilize the backward differential formula (BDF). The resulting
 202 first- and second-order temporal discretizations are shown as follows:

203 • **First-order backward Euler scheme**

$$\mathbf{q}^{n+1} = \mathbf{D} \nabla_{\mathbf{x}} \rho^{n+1}, \quad (3.7a)$$

$$\frac{\rho^{n+1} - \rho^n}{\Delta t} + e^{-\Delta t / \varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} \mathcal{S}_{\mathbf{v}}^n[g] \rangle = (1 - e^{-\Delta t / \varepsilon^2}) \nabla_{\mathbf{x}} \cdot \mathbf{q}^{n+1}, \quad (3.7b)$$

204 • **Second-order BDF scheme**

$$\mathbf{q}^{n+1} = \mathbf{D} \nabla_{\mathbf{x}} \rho^{n+1}, \quad (3.8a)$$

$$\frac{3\rho^{n+1} - 4\rho^n + \rho^{n-1}}{2\Delta t} + e^{-\Delta t / \varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} \mathcal{S}_{\mathbf{v}}^n[g] \rangle = (1 - e^{-\Delta t / \varepsilon^2}) \nabla_{\mathbf{x}} \cdot \mathbf{q}^{n+1}. \quad (3.8b)$$

205 In space, we take a DG approximation for the convection term and an LDG approximation
 206 for the diffusion term. A fully discrete scheme with first-order temporal accuracy is given as
 207 follows. Given $f_h^n \in V_h^k$, $\rho_h^n \in U_h^k$ and $g_h^n = (f_h^n - \rho_h^n)/\varepsilon$, we look for $\rho_h^{n+1} \in U_h^k$, $\mathbf{q}_h^{n+1} \in (U_h^k)^d$
 208 such that for any $K \in \mathcal{T}_h$, $\varphi \in U_h^k$, $\phi \in U_h^k$,

$$(\mathbf{q}_h^{n+1}, \varphi)_K = \mathbf{D} r_K(\rho_h^{n+1}, \varphi), \quad (3.9a)$$

$$\left(\frac{\rho_h^{n+1} - \rho_h^n}{\Delta t}, \phi \right)_K + e^{-\Delta t/\varepsilon^2} b_K(\langle \mathbf{v} \mathcal{S}_v^n[g_h] \rangle, \phi) = (1 - e^{-\Delta t/\varepsilon^2}) l_K(\mathbf{q}_h^{n+1}, \phi). \quad (3.9b)$$

209 Here $(\phi, \varphi)_K = \int_K \phi(\mathbf{x}) \varphi(\mathbf{x}) d\mathbf{x}$ and the bilinear forms in the scheme are

$$r_K(\rho, \varphi) = \sum_{e \in \partial K} \int_e \mathbf{n}_{e,K} \check{\rho}_{e,K} \varphi d\Gamma - \int_K \rho \nabla_{\mathbf{x}} \varphi d\mathbf{x}, \quad (3.10a)$$

$$l_K(\mathbf{q}, \phi) = \sum_{e \in \partial K} \int_e \mathbf{n}_{e,K} \cdot \hat{\mathbf{q}}_{e,K} \phi d\Gamma - \int_K \mathbf{q} \nabla_{\mathbf{x}} \phi d\mathbf{x}, \quad (3.10b)$$

$$b_K(\langle \mathbf{v} \mathcal{S}_v^n[g] \rangle, \phi) = \sum_{e \in \partial K} \int_e \langle \widetilde{\mathbf{v} \mathcal{S}_v^n[g]} \rangle_{e,K} \cdot \mathbf{n}_{e,K} \phi d\Gamma - \int_K \langle \mathbf{v} \mathcal{S}_v^n[g] \rangle \cdot \nabla_{\mathbf{x}} \phi d\mathbf{x}, \quad (3.10c)$$

210 where $\mathbf{n}_{e,K}$ is a unit outer normal vector to the edge e from the element K . $\check{\rho}_{e,K}^{(i)}$, $\hat{\mathbf{q}}_{e,K}^{(i)}$, $\widetilde{\mathbf{v} \mathcal{S}_v^n[g]}_{e,K}$
 211 are numerical fluxes. The integrals in (3.10c) are the most difficult parts that need very careful
 212 treatments. In short, $\mathcal{S}_v^n[g]$ is reconstructed on the upstream element K^* obtained by tracking
 213 an element K , while the test function ϕ is defined on K . We will discuss these integrals
 214 afterward. For the convection operator $\mathbf{v} \mathcal{S}_v^n[g]$, we use an upwind flux which is defined as

$$(\widetilde{\mathbf{v} \mathcal{S}_v^n[g]})_{e,K}(\mathbf{x}) = \mathbf{v} \{ \mathcal{S}_v^n[g] \}_{e,K} - \frac{|\mathbf{v}|}{2} [\mathcal{S}_v^n[g]]_{e,K} = \begin{cases} \mathbf{v} \mathcal{S}_v^n[g](\mathbf{x}^+), & \mathbf{v} \cdot \mathbf{n}_{e,K} > 0; \\ \mathbf{v} \mathcal{S}_v^n[g](\mathbf{x}^-), & \mathbf{v} \cdot \mathbf{n}_{e,K} < 0. \end{cases} \quad (3.11)$$

215 The pairs $(\check{\rho}, \hat{\mathbf{q}})$ are consistent numerical fluxes related to the diffusion operator. Here, we can
 216 take either of the following fluxes:

$$\text{alternating left-right: } \check{\rho}(\mathbf{x}) = \rho(\mathbf{x}^-), \quad \hat{\mathbf{q}}(\mathbf{x}) = \mathbf{q}(\mathbf{x}^+); \quad (3.12a)$$

$$\text{alternating right-left: } \check{\rho}(\mathbf{x}) = \rho(\mathbf{x}^+), \quad \hat{\mathbf{q}}(\mathbf{x}) = \mathbf{q}(\mathbf{x}^-), \quad (3.12b)$$

$$\text{central: } \check{\rho}(\mathbf{x}) = \frac{1}{2}(\rho(\mathbf{x}^-) + \rho(\mathbf{x}^+)), \quad \hat{\mathbf{q}}(\mathbf{x}) = \frac{1}{2}(\mathbf{q}(\mathbf{x}^-) + \mathbf{q}(\mathbf{x}^+)). \quad (3.12c)$$

217 In our numerical experiments, the alternating left-right flux is used.

218 Similarly, the fully discrete scheme with second-order temporal accuracy is shown here. The
 219 sole distinction from (3.9) lies in the first term on the left-hand side of (3.13b). Given $f_h^n \in V_h^k$,
 220 $\rho_h^{n-1} \in U_h^k$, $\rho_h^n \in U_h^k$ and $g_h^n = (f_h^n - \rho_h^n)/\varepsilon$, we look for $\rho_h^{n+1} \in U_h^k$, $\mathbf{q}_h^{n+1} \in (U_h^k)^d$ such that for
 221 any $K \in \mathcal{T}_h$, $\varphi \in U_h^k$, $\phi \in U_h^k$,

$$(\mathbf{q}_h^{n+1}, \varphi)_K = \mathbf{D} r_K(\rho_h^{n+1}, \varphi), \quad (3.13a)$$

$$\left(\frac{3\rho_h^{n+1} - 4\rho_h^n + \rho_h^{n-1}}{2\Delta t}, \phi \right)_K + e^{-\Delta t/\varepsilon^2} b_K(\langle \mathbf{v} \mathcal{S}_v^n[g_h] \rangle, \phi) = (1 - e^{-\Delta t/\varepsilon^2}) l_K(\mathbf{q}_h^{n+1}, \phi). \quad (3.13b)$$

222 3.2. An SL DG method for the microscopic equation

223 For the microscopic equation (2.19b), with the density ρ obtained from the macroscopic
 224 equation (2.19a), we can treat it as a transport type equation with a given source term. For
 225 this transport equation, we apply a backward SL DG scheme by following the idea in [48, 41].

226 We denote $K(t)$ as a dynamic element associated with Eulerian element K at time level
 227 t whose trajectory is bounded by characteristic lines. For example, if K is a 2D rectangle
 228 with nodes $\{\hat{\mathbf{x}}_i\}_{i=1}^4$, due to a constant velocity $\mathbf{v}/\varepsilon$, $K(t)$ is also a 2D rectangle with nodes
 229 $\{\mathbf{x}_i(t)\}_{i=1}^4$, where $\mathbf{x}_i(t) = \hat{\mathbf{x}}_i - (t^{n+1} - t)\mathbf{v}/\varepsilon$. Here we put K on the time level t^{n+1} , namely,
 230 $K^{n+1} = K(t^{n+1}) = K$, and denote $K^n = K(t^n) = K^*$ for the snapshot at time level t^n . We
 231 define a time dependent test function $\Psi(t, \mathbf{x})$ by the following adjoint final-value problem:

$$\partial_t \Psi + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} \Psi = 0, \quad \Psi(t^{n+1}, \mathbf{x}) = \psi(\mathbf{x}), \quad \forall \psi(\mathbf{x}) \in U_h^k. \quad (3.14)$$

232 Solving (3.14) by utilizing the method of characteristics gives

$$\Psi(t, \mathbf{x}) = \psi(\mathbf{x} - \mathbf{v}(t^{n+1} - t)/\varepsilon), \quad \forall t \in [t^n, t^{n+1}]. \quad (3.15)$$

233 In particular, the test functions at time level t^{n+1} and t^n are

$$\Psi^{n+1}(\mathbf{x}) = \Psi(t^{n+1}, \mathbf{x}) = \psi(\mathbf{x}) \quad \text{and} \quad \Psi^n(\mathbf{x}) = \Psi(t^n, \mathbf{x}) = \psi(\mathbf{x} - \mathbf{v}\Delta t/\varepsilon). \quad (3.16)$$

234 Applying Reynold's transport theorem and divergence theorem, with (3.14) and (2.19b), we
 235 obtain a weak formulation:

$$\begin{aligned} \frac{d}{dt} \int_{K(t)} f \Psi \, d\mathbf{x} &= \int_{K(t)} \partial_t(f \Psi) \, d\mathbf{x} + \int_{\partial K(t)} \frac{1}{\varepsilon} (\mathbf{v} \cdot \mathbf{n}) f \Psi \, d\Gamma \\ &= \int_{K(t)} \partial_t(f \Psi) \, d\mathbf{x} + \int_{\partial K(t)} \frac{1}{\varepsilon} (\mathbf{v} f \Psi) \cdot \mathbf{n} \, d\Gamma \\ &= \int_{K(t)} \partial_t(f \Psi) \, d\mathbf{x} + \int_{K(t)} \frac{1}{\varepsilon} \nabla_{\mathbf{x}} \cdot (\mathbf{v} f \Psi) \, d\mathbf{x} \\ &= \int_{K(t)} (\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f) \Psi + (\partial_t \Psi + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} \Psi) f \, d\mathbf{x} \\ &= \int_{K(t)} \frac{1}{\varepsilon^2} (\rho - f) \Psi \, d\mathbf{x}. \end{aligned} \quad (3.17)$$

236 Inside the time interval $[t^n, t^{n+1}]$, the last equality of (3.17) can be written in an integral form:
 237

$$\int_{K^{n+1}} f^{n+1} \Psi^{n+1} \, d\mathbf{x} = \int_{K^n} f^n \Psi^n \, d\mathbf{x} + \int_{t^n}^{t^{n+1}} \int_{K(t)} \frac{1}{\varepsilon^2} (\rho - f) \Psi \, d\mathbf{x}. \quad (3.18)$$

238 For the integral equation (3.18), in order to avoid the strong stiffness from the source term,
 239 we can take an implicit time discretization for the source term. For a first-order approximation,
 240 a backward Euler scheme is used. For a second-order approximation, a trapezoidal quadrature
 241 rule is applied. In space, we follow the standard DG discretizations. As a result, the first-order
 242 and second-order SL DG schemes for (3.18) are defined as follows. Given $f_h^n \in V_h^k$ and $\rho_h^k \in U_h^k$,
 243 we look for $f_h^{n+1} \in V_h^k$, such that for all $K \in \mathcal{T}_h$ and $\psi \in U_h^k$,

244 • **First-order backward SL DG scheme:**

$$(f_h^{n+1}, \psi)_K = \frac{\varepsilon^2}{\varepsilon^2 + \Delta t} a_K(f_h^n, \psi; \mathbf{v}) + \frac{\Delta t}{\varepsilon^2 + \Delta t} (\rho_h^{n+1}, \psi)_K; \quad (3.19)$$

245 • **Second-order Crank-Nicolson SL DG scheme:**

$$(f_h^{n+1}, \psi)_K = \frac{\varepsilon^2}{2\varepsilon^2 + \Delta t} a_K(f_h^n, \psi; \mathbf{v}) + \frac{\Delta t}{2\varepsilon^2 + \Delta t} (\rho_h^{n+1}, \psi)_K - \frac{\Delta t}{2\varepsilon^2 + \Delta t} a_K(f_h^n - \rho_h^n, \psi; \mathbf{v}). \quad (3.20)$$

246 Here the bilinear form $a_K(f, \psi; \mathbf{v})$, which depends on the velocity $\mathbf{v}$, is defined as:

$$a_K(f, \psi; \mathbf{v}) = \int_{K^*} f(\mathbf{x}^*) \psi(\mathbf{x}^* + \mathbf{v} \Delta t / \varepsilon) d\mathbf{x}^* = \int_K f(\mathbf{x} - \mathbf{v} \Delta t / \varepsilon) \psi(\mathbf{x}) d\mathbf{x} = \int_K \mathcal{S}_v^n[f] \psi d\mathbf{x}. \quad (3.21)$$

247 We note that for second order, we take a Crank-Nicolson scheme, rather a second order multi-
 248 step scheme as in (3.13) for ρ . The idea is to avoid the characteristic tracking across two time
 249 levels which may make the computation of $a_K(f, \psi; \mathbf{v})$ in (3.21) more complicate and more
 250 costly.

251 *3.3. An oscillation-free damping approach*

252 We now address the issue of numerical oscillations that may arise when high-order methods
 253 are applied to non-smooth problems. To control spurious oscillations for implicit methods, we
 254 adopt the damping technique [43, 44] specifically for the f equations. Note that we do not
 255 apply a damping approach to the ρ equation, because at each step the real density ρ will be
 256 determined by averaging f in velocity, see the next section for more details. This damping is to
 257 suppress high-order moments of a DG scheme in a local way by using a coefficient depending
 258 on the smoothness of the solution. However the zeroth moment is not affected so that mass
 259 conservation is preserved. In our method, let us replace f with $f_h \in V_h^k$ in (3.17), a semi-discrete
 260 scheme which includes the damping term is defined as follows,

$$\frac{d}{dt} \int_{K(t)} f_h \Psi d\mathbf{x} = \int_{K(t)} \frac{1}{\varepsilon^2} (\rho_h - f_h) \Psi d\mathbf{x} - \sum_{\ell=0}^k \frac{\sigma_K^\ell(t)}{h_K} \int_{K(t)} (f_h - \pi_h^{\ell-1} f_h) \Psi d\mathbf{x}, \quad (3.22)$$

261 where π_h^ℓ ($\ell \geq 0$) is a standard $L^2(\Omega_x)$ projection onto V_h^ℓ , that is, for any function w , $\pi_h^\ell w \in V_h^\ell$
 262 satisfies

$$\int_K (\pi_h^\ell w - w) v_h d\mathbf{x} = 0, \quad \forall v_h \in \mathcal{P}^\ell(K). \quad (3.23)$$

263 We also define $\pi_h^{-1} = \pi_h^0$. The damping coefficient is defined as

$$\sigma_K^\ell(t) = \frac{2(2\ell+1)}{2k-1} \frac{h^\ell}{\ell!} \sum_{|\alpha|=\ell} \left(\frac{1}{N_e} \llbracket \partial^\alpha f_h(t) \rrbracket^2 \right)^{\frac{1}{2}}. \quad (3.24)$$

264 In implementation, we take the damping term at the time level t^{n+1} but approximate $\sigma_K^\ell(t)$
 265 using the information at the previous time level t^n . In this case, the damping term is linear
 266 with respect to the unknown f_h while the temporal accuracy is still kept due to the projection
 267 as shown in (3.23). In what follows, we denote $\sigma_K^{\ell,n} = \sigma_K^\ell(t^n)$ for simplicity.

268 We take the second-order SL DG scheme (3.20) in the modal form as an example here.
 269 Assume that $\{\psi_\ell(\mathbf{x})\}_{\ell=0}^k \subset \mathcal{P}^k(K)$ is a set of orthonormal basis functions such that

$$\int_K \psi_\ell(\mathbf{x}) \psi_m(\mathbf{x}) d\mathbf{x} = \delta_{\ell m}, \quad \text{for all } 0 \leq \ell, m \leq k, \quad (3.25)$$

270 where $\delta_{\ell m}$ is the Kronecker delta. For $\mathbf{x} \in K$, we have

$$f_h^n(\mathbf{x}) = \sum_{\ell=0}^k f_\ell^n \psi_\ell(\mathbf{x}), \quad \rho_h^n(\mathbf{x}) = \sum_{\ell=0}^k \rho_\ell^n \psi_\ell(\mathbf{x}). \quad (3.26)$$

271 Substitute (3.26) into (3.20), we obtain that

$$f_m^{n+1} = \frac{\varepsilon^2 a_{m,1}^n + \Delta t \rho_m^{n+1} - \Delta t a_{m,2}^n}{2\varepsilon^2 + \Delta t} - \frac{2\varepsilon^2 \Delta t}{2\varepsilon^2 + \Delta t} \sum_{\ell=0}^k \frac{\sigma_K^{\ell,n}}{h_K} \sum_{j=\max(1,\ell)}^k f_j^{n+1} \delta_{jm}, \quad (3.27)$$

272 where $a_{m,1}^n, a_{m,2}^n$ correspond to the two $a_K(\cdot, \cdot)$ terms in (3.21). The first term on the right-hand
 273 side of (3.27) represents the original scheme, while the second term is the damping term. As
 274 we observe when $m = 0$, the damping term is zero, so the 0-th moment is unchanged compared
 275 to the original scheme. Namely, the mass conservation is not affected. When $m \geq 1$, we have

$$\begin{aligned} \sum_{\ell=0}^k \frac{\sigma_K^{\ell,n}}{h_K} \sum_{j=\max(1,\ell)}^k f_j^{n+1} \delta_{jm} &= \frac{\sigma_K^{0,n}}{h_K} f_m^{n+1} + \sum_{\ell=1}^k \sum_{j=\ell}^k \frac{\sigma_K^{\ell,n}}{h_K} f_j^{n+1} \delta_{jm} \\ &= \frac{\sigma_K^{0,n}}{h_K} f_m^{n+1} + \sum_{j=1}^k \sum_{\ell=1}^j \frac{\sigma_K^{\ell,n}}{h_K} f_j^{n+1} \delta_{jm} = \sum_{\ell=0}^m \frac{\sigma_K^{\ell,n}}{h_K} f_m^{n+1}. \end{aligned} \quad (3.28)$$

276 Let $b_0^n = 0$ and $b_m^n = \sum_{\ell=0}^m \frac{\sigma_K^{\ell,n}}{h_K}$ for $1 \leq m \leq k$, we obtain the updating formula for f_m^{n+1} ,

$$f_m^{n+1} = \frac{\varepsilon^2 a_{m,1}^n + \Delta t \rho_m^{n+1} - \Delta t a_{m,2}^n}{2\varepsilon^2 + (1 + 2\varepsilon^2 b_m^n) \Delta t}, \quad \forall 0 \leq m \leq k. \quad (3.29)$$

277 As we can see, the damping term is linear with respect to the unknown f_m^{n+1} , so that it
 278 is very convenient for implicit schemes, as compared to TVB limiters in [27]. Besides, it does
 279 not bring too much computational cost as compared to the original scheme. Numerical tests
 280 in Section 5 show that this damping approach can properly control numerical oscillations.

281 3.4. Overall algorithm

282 We now summarize the overall algorithm by collecting all the above proposed ingredients.
 283 We name our method SL-LDG- p - q ($p \leq 2$), representing a p -th order temporal approximation
 284 and a q -th order spatial approximation. If $p = q$, we take SL-LDG- p for short. In velocity, we
 285 will apply the discrete velocity method [49]. Let $\{\mathbf{v}_\alpha\}_{\alpha=1}^{N_v}$ denote a set of quadrature points in Ω_v
 286 and $\{w_\alpha\}_{\alpha=1}^{N_v}$ be the corresponding quadrature weights, where N_v is the number of quadrature
 287 points. An integral in velocity is approximated by

$$\langle f(\mathbf{v}) \rangle = \int_{\Omega_v} f(\mathbf{v}) d\mu \approx \sum_{\alpha=1}^{N_v} w_\alpha f(\mathbf{v}_\alpha). \quad (3.30)$$

288 To simplify the algorithm description in the following, we keep the notation $\langle \cdot \rangle$ but use the
 289 above approximation of integral in velocity in practice. Additionally, in order to improve the
 290 robustness, inspired from [22, 45], we will enforce $\langle g \rangle = 0$ by introducing a predictor-corrector
 291 procedure.

292 The **SL-LDG-1- q** method is described as follows. Let $k = q - 1$. At $t = 0$, the numerical
 293 solutions are initialized via projections:

$$f_h^0(\mathbf{x}, \mathbf{v}) = \pi_h^k(f^0), \quad \rho_h^0(\mathbf{x}) = \langle f_h^0(\mathbf{x}, \mathbf{v}) \rangle, \quad g_h^0(\mathbf{x}, \mathbf{v}) = (f_h^0(\mathbf{x}, \mathbf{v}) - \rho_h^0(\mathbf{x}))/\varepsilon. \quad (3.31)$$

294 At time level t^n , we look for $f_h^{n+1} \in V_h^k$, $\rho_h^{n+1} \in U_h^k$, such that for any $K \in \mathcal{T}_h$, $\varphi, \phi, \gamma \in U_h^k$,
 295 $\psi, \zeta \in V_h^k$,

$$(\tilde{\mathbf{q}}_h^{n+1}, \varphi)_K = D r_K(\tilde{\rho}_h^{n+1}, \varphi), \quad (3.32a)$$

$$\left(\frac{\tilde{\rho}_h^{n+1} - \rho_h^n}{\Delta t}, \phi \right)_K + e^{-\Delta t/\varepsilon^2} b_K(\langle \mathbf{v} \mathcal{S}_v^n[g_h] \rangle, \phi) = (1 - e^{-\Delta t/\varepsilon^2}) l_K(\tilde{\mathbf{q}}_h^{n+1}, \phi), \quad (3.32b)$$

$$(f_h^{n+1}, \psi)_K = \frac{\varepsilon^2}{\varepsilon^2 + \Delta t} a_K(f_h^n, \psi; \mathbf{v}) + \frac{\Delta t}{\varepsilon^2 + \Delta t} (\tilde{\rho}_h^{n+1}, \psi)_K, \quad (3.32c)$$

$$(\rho_h^{n+1}, \gamma)_K = (\langle f_h^{n+1} \rangle, \gamma)_K, \quad (3.32d)$$

$$(g_h^{n+1}, \zeta)_K = ((f_h^{n+1} - \rho_h^{n+1})/\varepsilon, \zeta)_K. \quad (3.32e)$$

296 The bilinear forms are provided in (3.10) in Section 3.1 and (3.21) in Section 3.2. The temporary
 297 variable g_h^n introduced here can be manually eliminated using (3.32e). To implement this
 298 method, we first substitute (3.32a) into (3.32b) to form a low-dimensional (space-only) sparse
 299 linear algebraic system. By solving such a system, we get a macroscopic prediction $\tilde{\rho}_h^{n+1}$ at
 300 time level t^{n+1} . Based on this, the high-dimensional unknowns f_h^{n+1} at all discrete velocities are
 301 completely decoupled. Thereby we are able to efficiently compute f_h^{n+1} by (3.32c) via parallel
 302 computing. We finish the algorithm by correcting the macroscopic and microscopic unknowns
 303 ρ_h^{n+1} , g_h^{n+1} at t^{n+1} using projections (3.32d) and (3.32e) to keep an exact conservation of mass.

304 The **SL-LDG-2- q** method is almost the same. At time level t^n , we look for $f_h^{n+1} \in V_h^k$,
 305 $\rho_h^{n+1} \in U_h^k$, such that for all $K \in \mathcal{T}_h$, $\varphi, \phi, \gamma \in U_h^k$, $\psi, \zeta \in V_h^k$,

$$(\tilde{\mathbf{q}}_h^{n+1}, \varphi)_K = D r_K(\tilde{\rho}_h^{n+1}, \varphi), \quad (3.33a)$$

$$\left(\frac{3\tilde{\rho}_h^{n+1} - 4\rho_h^n + \rho_h^{n-1}}{2\Delta t}, \phi \right)_K + e^{-\Delta t/\varepsilon^2} b_K(\langle \mathbf{v} \mathcal{S}_v^n[g_h] \rangle, \phi) = (1 - e^{-\Delta t/\varepsilon^2}) l_K(\tilde{\mathbf{q}}_h^{n+1}, \phi), \quad (3.33b)$$

$$(f_h^{n+1}, \psi)_K = \frac{\varepsilon^2}{2\varepsilon^2 + \Delta t} a_K(f_h^n, \psi; \mathbf{v}) + \frac{\Delta t}{2\varepsilon^2 + \Delta t} (\tilde{\rho}_h^{n+1}, \psi)_K - \frac{\Delta t}{2\varepsilon^2 + \Delta t} a_K((f_h^n - \rho_h^n), \psi; \mathbf{v}), \quad (3.33c)$$

$$(\rho_h^{n+1}, \gamma)_K = (\langle f_h^{n+1} \rangle, \gamma)_K, \quad (3.33d)$$

$$(g_h^{n+1}, \zeta)_K = ((f_h^{n+1} - \rho_h^{n+1})/\varepsilon, \zeta)_K. \quad (3.33e)$$

306 To start this multi-step solver, we could apply the **SL-LDG-1- q** method (3.32) for $n = 0$ and
 307 then run with (3.33) for $n \geq 1$.

308 **Remark 3.1.** To implement the **proposed** method, we have to evaluate four types of integrals:

309 (a) $\int_K u(\mathbf{x})v(\mathbf{x})d\mathbf{x}$, e.g., volume integrals in $r_K(\cdot, \cdot)$, $l_K(\cdot, \cdot)$;

310 (b) $\int_{\partial K} u(\mathbf{x})v(\mathbf{x})d\mathbf{x}$, e.g., flux integrals in $r_K(\cdot, \cdot)$, $l_K(\cdot, \cdot)$;

311 (c) $\int_K u(\mathbf{x} - a\mathbf{y})v(\mathbf{x})d\mathbf{x}$, e.g., volume integrals in $a_K(\cdot, \cdot; \mathbf{v})$, $b_K(\cdot, \cdot)$;

312 (d) $\int_{\partial K} u(\mathbf{x} - a\mathbf{y})v(\mathbf{x})d\mathbf{x}$, e.g., flux integrals in $b_K(\cdot, \cdot)$.

313 For simplicity, we assume that K is rectangular. To evaluate (a) and (b), we can simply apply
 314 a Gauss-Legendre quadrature. However, evaluating integrals (c) and (d) is more challenging
 315 as $u(\mathbf{x} - a\mathbf{y})$ might cross elements. Our approach for handling integrals (c) and (d) comprises
 316 three steps: (1) tracking elements along characteristic lines, (2) clipping upstream elements
 317 with background elements, and (3) collocating quadrature points at the intersection region.
 318 Thanks to constant velocities, upstream elements are rectangular. Let us take a 2D uniform
 319 mesh as an example. As depicted in Figure 3.1, an element K can be clipped into four small
 320 sub-elements $\{K_\alpha\}_{\alpha=1}^4$, ensuring continuity of $u(\mathbf{x} - a\mathbf{y})$ within each sub-element. To evaluate
 321 integrals of type (c), we assign 2D Gauss-Legendre points (black crosses) to each subregion and
 322 compute them as follows:

$$\int_K u(\mathbf{x} - a\mathbf{y})v(\mathbf{x})d\mathbf{x} = \sum_{\alpha} \int_{K_{\alpha}} u(\mathbf{x} - a\mathbf{y})v(\mathbf{x})d\mathbf{x} \approx \sum_{\alpha, \beta} w_{\beta} u(\mathbf{x}_{\alpha, \beta}^*) v(\mathbf{x}_{\alpha, \beta}), \quad (3.34)$$

323 where $\mathbf{x}_{\alpha, \beta}^*$ corresponds to the black cross on the left and $\mathbf{x}_{\alpha, \beta}$ on the right. Similarly, for each
 324 interface ∂K (or ∂K^*), we apply a 1D Gauss-Legendre quadrature (blue stars) to compute
 325 line integrals of type (d). It is worth noting that the key aspect of the clipping algorithm for
 326 a hyperrectangle is locating points of intersection within an element, namely, the central red
 327 circles in Figure 3.1. As we can see here, with constant velocities, clipping can be achieved
 328 using 1D algorithms through dimensional splitting, which makes it easy to extend our method
 to 3D cases.

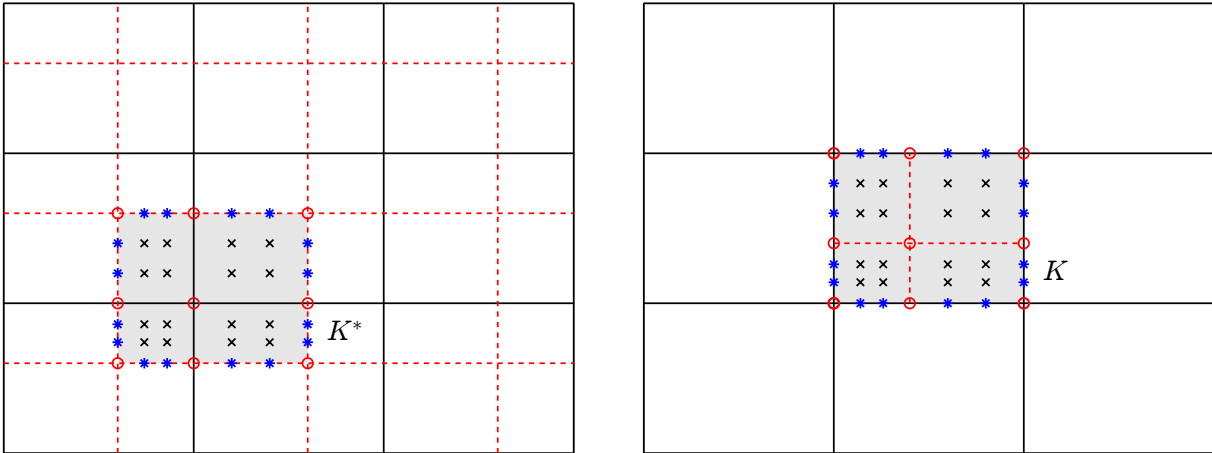


Figure 3.1: Illustration of quadrature points for convolution integrals: Lagrangian element K^* (left); Eulerian element K (right). The black solid lines represent Eulerian mesh, while the red dashed lines represent Lagrangian mesh. The red circles determine the geometric partitions. The black crosses denote the Gauss-Legendre points for the volume integral of type (c), while the blue stars are used for the flux integral of type (d).

329
 330 **Remark 3.2.** Here we briefly discuss the boundary treatment; for more details, we refer to
 331 [12, 13, 23]. By using an SL method, we track a point from $(t, \mathbf{x})$ to $(t_*, \mathbf{x}_*)$. When the point
 332 $\mathbf{x}_*$ exceeds the computational domain $\Omega_{\mathbf{x}}$, we need to assign ghost values for $f(t_*, \mathbf{x}_*, \mathbf{v})$ and

333 $\rho(t_*, x_*)$. Taking the 1D case as an example, we focus on the left boundary. For a periodic
 334 boundary condition, let P be a periodic mapping that maps a ghost point x_* to an interior
 335 point $P(x_*)$, as shown in Figure 3.2(a). We simply set

$$f(t_*, x_*, v) = f(t_*, P(x_*), v) \quad \text{and} \quad \rho(t_*, x_*) = \rho(t_*, P(x_*)). \quad (3.35)$$

336 For a Dirichlet inflow boundary condition, the characteristics would intersect with the left
 337 boundary at $(\tilde{t}, x_L)$ as shown in Figure 3.2(b). In this case, we take a first-order approximation
 338 by using the values at (t_*, x_L) to approximate the values at $(\tilde{t}, x_L)$, that is

$$f(t_*, x_*, v) = f_L(\tilde{t}, v) \approx f_L(t_*, v) \quad \text{and} \quad \rho(t_*, x_*) = \rho_L(\tilde{t}) \approx \rho_L(t_*). \quad (3.36)$$

339 We note that $\rho_L(t_*)$ is obtained by using a close-loop boundary condition as detailed in [12, 13,
 340 23, 27].

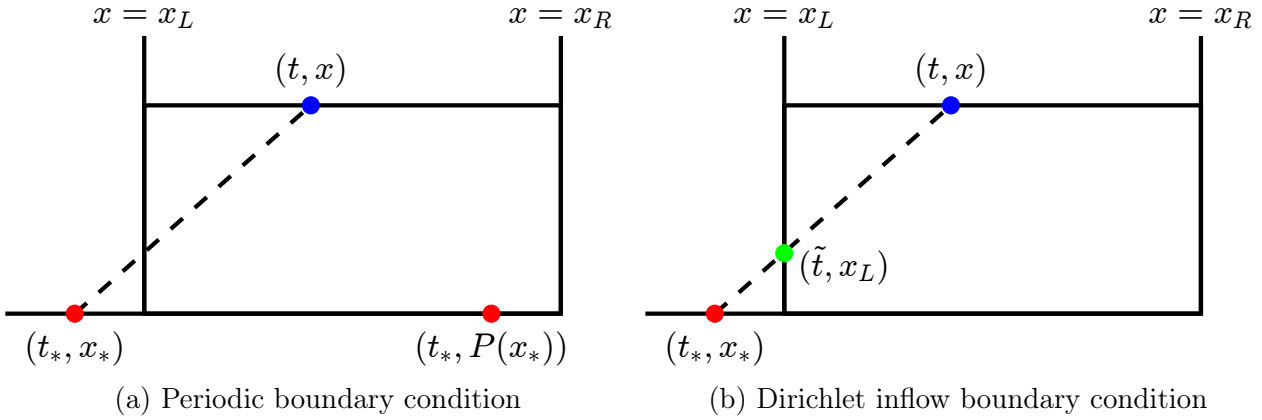


Figure 3.2: Graphic description of boundary treatment for exterior points.

341 4. Theoretical Analysis

342 In this section, we first analyze the AP property of the proposed **SL-LDG- p - q** methods by
 343 using the simple linear telegraph equation (2.1a). Then we conduct a Fourier analysis to study
 344 their stability as in [27].

345 4.1. Formal asymptotic analysis

346 **Theorem 4.1.** Assume that the telegraph equation (1.1)(2.1a) is equipped with a periodic bound-
 347 ary condition and a well-prepared initial condition. For a fixed time step Δt and mesh size h ,
 348 as $\varepsilon \rightarrow 0$, the **SL-LDG- p - q** methods ($p = 1, 2, q \geq 1$) are formally AP.

349 *Proof.* We only consider the **SL-LDG-1- q** method for the telegraph equation. The proof of the
 350 **SL-LDG-2- q** method is almost the same. Let $\varepsilon \rightarrow 0$, (3.32) formally becomes

$$\begin{aligned} (q_h^{n+1}, \varphi)_K &= r_K(\tilde{\rho}_h^{n+1}, \varphi), \quad \forall \varphi \in U_h^k; \\ (\frac{\tilde{\rho}_h^{n+1} - \rho_h^n}{\Delta t}, \phi)_K &= l_K(q_h^{n+1}, \phi), \quad \forall \phi \in U_h^k; \\ (f_h^{n+1}, \psi)_K &= (\tilde{\rho}_h^{n+1}, \psi)_K, \quad \forall \psi \in V_h^k; \\ (\rho_h^{n+1}, \gamma)_K &= (\langle f_h^{n+1} \rangle, \gamma), \quad \forall \gamma \in V_h^k. \end{aligned}$$

351 This implies $\rho_h^{n+1} = \langle f_h^{n+1} \rangle = f_h^{n+1} = \tilde{\rho}_h^{n+1}$, where $\tilde{\rho}_h^{n+1}$ satisfies a backward Euler LDG scheme
 352 for the limiting heat equation. $\square$

353 4.2. Fourier stability analysis

354 Here we carry out a Fourier analysis in the one-dimensional case. We employ a periodic
 355 boundary condition and a uniform mesh with cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ for $1 \leq i \leq N$. Besides, we
 356 consider the **SL-LDG- p - q** methods with $p = 1, 2$, $q \geq 1$ and the left-right alternating flux pair,
 357 applied to the telegraph equation where $\Omega_v = \{-1, 1\}$. Let the numerical solutions be

$$\rho_h^n(x) = \sum_{l=0}^{q-1} \rho_{i,l}^n \phi_i^l(x), \quad q_h^n(x) = \sum_{l=0}^{q-1} q_{i,l}^n \phi_i^l(x), \quad f_h^n(x, v_\alpha) = \sum_{l=0}^{q-1} f_{i,l,\alpha}^n \phi_i^l(x), \quad \forall x \in I_i, \quad (4.1)$$

358 where $v_\alpha = -1, 1$, $\phi_i^l(x) = \phi^l(\xi_i)$ with $\xi_i = \frac{x-x_i}{h_i}$ and ϕ^l is the l -th scaled Legendre polynomial
 359 on $[-\frac{1}{2}, \frac{1}{2}]$,

$$\phi^0(\xi) = 1, \quad \phi^1(\xi) = \xi, \quad \phi^2(\xi) = \xi^2 - \frac{1}{12}, \quad \dots \quad (4.2)$$

360 We collect unknowns as vectors $\boldsymbol{\rho}_i^n = (\rho_{i,0}^n, \dots, \rho_{i,q-1}^n)^T$, $\boldsymbol{f}_{i,\alpha}^n = (f_{i,0,\alpha}^n, \dots, f_{i,q-1,\alpha}^n)^T$ and
 361 take the ansatz $\boldsymbol{\rho}_i^n = \hat{\boldsymbol{\rho}}^n \exp(\mathcal{I}\kappa x_i)$, $\boldsymbol{f}_{i,\alpha}^n = \hat{\boldsymbol{f}}_\alpha^n \exp(\mathcal{I}\kappa x_i)$ with the imaginary unit $\mathcal{I}^2 = -1$,
 362 $\alpha = 1, \dots, N_v$. Note that the auxiliary variable q_h^n can be eliminated. Denoting

$$\boldsymbol{U}_1^n = (\hat{\boldsymbol{\rho}}^n, \hat{\boldsymbol{f}}_1^n, \dots, \hat{\boldsymbol{f}}_{N_v}^n, \hat{\boldsymbol{\rho}}^n)^T, \quad \boldsymbol{U}_2^n = (\hat{\boldsymbol{\rho}}^n, \hat{\boldsymbol{f}}_1^n, \dots, \hat{\boldsymbol{f}}_{N_v}^n, \hat{\boldsymbol{\rho}}^n, \hat{\boldsymbol{\rho}}^{n-1})^T,$$

363 the **SL-LDG- p - q** methods for telegraph equation can be written as

$$\boldsymbol{U}_p^{n+1} = \boldsymbol{A}^{p,q}(\varepsilon, h, \Delta t; \omega) \boldsymbol{U}_p^n, \quad (4.3)$$

364 where $\boldsymbol{A}^{p,q}$ is the amplification matrix depending on the model parameter ε , mesh size h , time
 365 step size Δt , and the discrete wave number $\omega = \kappa h \in [0, 2\pi]$. Theorem 4.2 illustrates the
 366 structure of the amplification matrix $\boldsymbol{A}^{p,q}$, with its proof presented in the appendix.

367 **Theorem 4.2.** *For any given $q \geq 1$ and $p = 1, 2$, the amplification matrix $\boldsymbol{A}^{p,q}$ related to*
 368 ***SL-LDG- p - q** method depends only on the terms $\frac{\varepsilon}{h}$ and $\frac{\Delta t}{\varepsilon h}$, namely, $\boldsymbol{A}^{p,q} = \boldsymbol{A}^{p,q}(\frac{\varepsilon}{h}, \frac{\Delta t}{\varepsilon h}; \omega)$.*

369 Next, we use the following principle to characterize the non-increasing L^2 energy stability.

370 **Principle for Numerical Stability [23].** For any given $\varepsilon, h, \Delta t$, let the eigenvalues of $\boldsymbol{A}$ be
 371 $\lambda_i(\omega)$, $i = 1, \dots, N$. Our scheme is “stable”, if for all $\omega \in [0, 2\pi]$, it satisfies either

$$\max_{i=1, \dots, N} \{|\lambda_i(\omega)|\} < 1, \quad \text{or} \quad (4.4a)$$

$$\max_{i=1, \dots, N} \{|\lambda_i(\omega)|\} = 1 \quad \text{and} \quad \boldsymbol{A} \text{ is diagonalizable.} \quad (4.4b)$$

372 We verify the uniformly unconditional stability of the proposed schemes from numerical
 373 sampling. Motivated by Theorem 4.2, we define $\sigma = \log_{10}(\varepsilon/h)$ and $\eta = \log_{10}(\varepsilon^2/\Delta t)$. Subse-
 374 quently, we uniformly sample σ using 1000 points within $[-5, 5]$, and η with 800 points inside
 375 $[-4, 4]$. Additionally, we sample the discrete wave number ω uniformly within $[0, 2\pi]$ using 500
 376 points. Based on the above stability principle, the **SL-LDG- p - q** methods with $p = 1, 2$ and
 377 $q = 1, 2$ are verified to be uniformly unconditionally stable.

378 5. Numerical Results

379 In this section, we present a set of numerical tests to demonstrate the performance of the
 380 proposed first-order and second-order **SL-LDG- p** ($p = 1, 2$) methods in terms of their accuracy,
 381 stability and efficiency, when applied to the multiscale problem (1.1) with either smooth or
 382 non-smooth solutions. For the second-order scheme, when applied to non-smooth problems,
 383 the oscillation-free damping limiter is used, and we denote the scheme as **SL-LDG-2-D**. The
 384 spatial domain is taken as $\Omega_{\mathbf{x}} = [a, b]^d$, with d being the dimension in space. For simplicity,
 385 a uniform spatial mesh with N^d elements is used and the mesh size is $h = (b - a)/N$. We
 386 apply the boundary treatment described in Remark 3.2. We take a time step $\Delta t = Ch$ where
 387 C is a referred CFL number, which is independent of the Knudsen number ε and mesh size
 388 h , but would be specific to each problem. By using an SL method with DG discretization
 389 for the convection part, the scheme is even more efficient than the finite difference scheme in
 390 [27]. Besides, we will show that the scheme is highly parallel, especially for high-dimensional
 391 problems (3D in $\mathbf{x}$ and 2D in $\mathbf{v}$).

392 5.1. One-dimensional transport problem

393 In this subsection, we consider 1D transport problems. We start with two-velocity models
 394 (2.1a)-(2.1b), and then turn to the one-group transport equation in a slab geometry (2.2) in
 395 the case $d = 1$. For each collision operator, the corresponding approximation model has been
 396 shown in [27].

397 5.1.1. Telegraph equation

398 To evaluate the performance of the proposed methods for the telegraph equation (2.1a), we
 399 present two representative tests here.

400 **Case 1: Accuracy test.** Consider the following smooth, exact solution

$$f(t, x, v) = \frac{1}{4} \left[-\frac{1}{\gamma} e^{-\gamma t} \sin(x) + v \varepsilon e^{-\gamma t} \cos(x) \right] + \frac{1}{2}, \quad \gamma = \frac{2}{1 + \sqrt{1 - 4\varepsilon^2}} \quad (5.1)$$

401 on $\Omega_x = [-\pi, \pi]$ with a periodic boundary condition. The numerical errors for the macroscopic
 402 density ρ in a L^r -norm and corresponding convergence orders at a final time T are defined as

$$E_{N,r}^\rho = \|\rho_h(T, x) - \rho(T, x)\|_{L^r}, \quad O_{N,r}^\rho = \log_2(E_{N,r}^\rho / E_{2N,r}^\rho), \quad (5.2)$$

403 for $r = 1, \infty$ respectively. We take $T = 1$ in this example. In Tables 5.1 and 5.2, we observe
 404 the desired convergence orders and uniform stability of p -th ($p = 1, 2$) order schemes in both
 405 kinetic and diffusive regimes. An order reduction occurs when $\varepsilon = \mathcal{O}(\sqrt{\Delta t})$, especially for the
 406 second-order scheme, which coincides with the theoretical results in Theorem 2.2 and numerical
 407 results of the finite difference scheme in [27].

408 **Case 2: Square wave.** Given a discontinuous square-shaped initial data from [12]:

$$f(x, v, 0) = 1 - \mathcal{H}(|x| - 0.2), \quad x \in [-1, 1] \quad (5.3)$$

409 with a Dirichlet boundary condition, where $\mathcal{H}(x)$ is the Heaviside function, we will show the
 410 effectiveness of the oscillation-free limiter with a damping in controlling numerical oscillations.
 411 In Figure 5.1, we show the numerical solutions at time $T = 0.5$ for $\varepsilon = 1.0$ and $T = 0.05$
 412 for $\varepsilon = 10^{-6}$. We take a fairly large time step $\Delta t = 1.5\Delta x$ with $h = 0.025$. In the kinetic
 413 regime, the original **SL-LDG-2** scheme exhibits clear spurious numerical oscillations, which can
 414 be significantly reduced by adding the damping term. Besides, the proposed methods are shown
 415 to be AP and stable with a large time step in the diffusive regime $\varepsilon = 10^{-6}$.

Table 5.1: Accuracy test for the telegraph equation (Section 5.1.1). Numerical errors and convergence orders of the **SL-LDG-1** method in L^1 and L^∞ norms. $T = 1$.

Parameters		$\Delta t = 0.5\Delta x$				$\Delta t = 5.0\Delta x$			
ε	N	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$
0.5	80	1.59E-02	-	4.48E-03	-	1.38E-01	-	3.46E-02	-
	160	7.61E-03	1.06	2.18E-03	1.04	8.01E-02	0.79	2.00E-02	0.79
	320	3.73E-03	1.03	1.08E-03	1.02	3.51E-02	1.19	8.80E-03	1.19
	640	1.85E-03	1.01	5.35E-04	1.01	1.42E-02	1.31	3.55E-03	1.31
10^{-1}	80	1.43E-02	-	4.68E-03	-	6.05E-02	-	1.54E-02	-
	160	9.11E-03	0.65	2.86E-03	0.71	3.72E-02	0.70	9.43E-03	0.71
	320	4.96E-03	0.88	1.63E-03	0.81	2.10E-02	0.83	5.30E-03	0.83
	640	2.37E-03	1.07	8.59E-04	0.93	1.30E-02	0.69	3.28E-03	0.69
10^{-2}	80	1.08E-02	-	4.05E-03	-	5.75E-02	-	1.47E-02	-
	160	5.39E-03	1.00	2.02E-03	1.00	3.32E-02	0.79	8.44E-03	0.80
	320	2.70E-03	1.00	1.01E-03	1.00	1.73E-02	0.94	4.38E-03	0.95
	640	1.36E-03	0.99	5.07E-04	1.00	8.85E-03	0.96	2.24E-03	0.97
10^{-6}	80	1.07E-02	-	4.05E-03	-	5.75E-02	-	1.47E-02	-
	160	5.36E-03	1.00	2.02E-03	1.00	3.32E-02	0.79	8.43E-03	0.80
	320	2.68E-03	1.00	1.01E-03	1.00	1.72E-02	0.94	4.37E-03	0.95
	640	1.34E-03	1.00	5.02E-04	1.00	8.81E-03	0.97	2.23E-03	0.97

Table 5.2: Accuracy test for the telegraph equation (Section 5.1.1). Numerical errors and convergence orders of the **SL-LDG-2** method in L^1 and L^∞ norms. $T = 1$.

Parameters		$\Delta t = 0.5\Delta x$				$\Delta t = 5.0\Delta x$			
ε	N	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$
0.5	80	2.37E-03	-	6.12E-04	-	1.04E-01	-	2.60E-02	-
	160	6.14E-04	1.95	1.58E-04	1.95	4.36E-02	1.25	1.09E-02	1.25
	320	1.56E-04	1.98	4.00E-05	1.98	1.34E-02	1.70	3.35E-03	1.70
	640	3.92E-05	1.99	1.01E-05	1.99	3.65E-03	1.87	9.13E-04	1.88
10^{-1}	80	4.04E-03	-	1.10E-03	-	3.49E-02	-	8.81E-03	-
	160	3.07E-03	0.40	7.91E-04	0.48	1.08E-02	1.68	2.73E-03	1.69
	320	1.53E-03	1.00	3.89E-04	1.02	5.21E-03	1.06	1.31E-03	1.06
	640	5.57E-04	1.46	1.41E-04	1.47	4.17E-03	0.32	1.04E-03	0.33
10^{-2}	80	2.81E-04	-	1.64E-04	-	3.15E-02	-	7.98E-03	-
	160	9.67E-05	1.54	4.75E-05	1.78	7.34E-03	2.10	1.86E-03	2.10
	320	5.17E-05	0.90	1.87E-05	1.34	1.66E-03	2.14	4.21E-04	2.14
	640	4.05E-05	0.35	1.16E-05	0.69	4.23E-04	1.97	1.07E-04	1.97
10^{-6}	80	2.45E-04	-	1.54E-04	-	3.15E-02	-	7.97E-03	-
	160	5.99E-05	2.03	3.83E-05	2.01	7.30E-03	2.11	1.85E-03	2.11
	320	1.49E-05	2.01	9.55E-06	2.00	1.62E-03	2.17	4.12E-04	2.17
	640	3.71E-06	2.00	2.38E-06	2.00	3.86E-04	2.07	9.79E-05	2.07

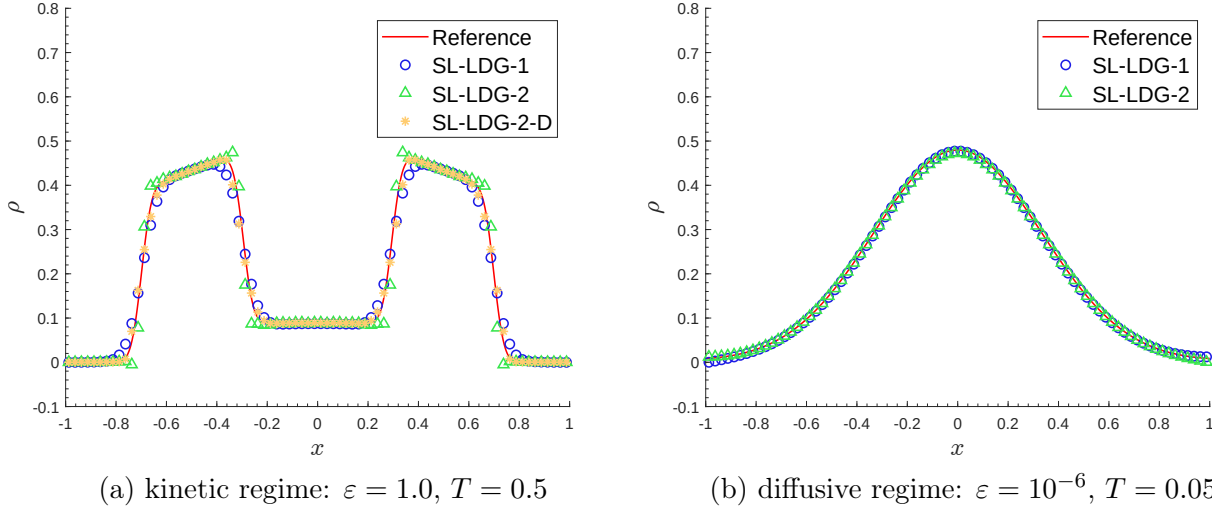


Figure 5.1: Square wave problem for the telegraph equation (Section 5.1.1). Density ρ of the **SL-LDG- p** ($p = 1, 2$) solutions at different regimes. $h = 0.025$ and $\Delta t = 1.5h$. The reference solution is computed by the first-order finite difference method based on micro-macro decomposition with $h = 0.004$ and $\Delta t = 0.001$.

5.1.2. Advection-diffusion equation

We now turn to the model with the collision operator (2.1b), which converges to an advection-diffusion equation as $\varepsilon \rightarrow 0$. Similarly, two cases are considered.

Case 1: Accuracy test in the diffusive regime. Let $A = 1$ and $\Omega_x = [-\pi, \pi]$, consider the following exact solution

$$f(t, x, v) = e^{-t} \sin(x - t) + \varepsilon v e^{-t} (\sin(x - t) - \cos(x - t)), \quad \rho(t, x) = e^{-t} \sin(x - t) \quad (5.4)$$

and a periodic boundary condition. We carry out a convergence study at $T = 1$ and $\varepsilon = 10^{-6}$. Table 5.3 shows the desired convergence behaviors of the first- and second-order schemes under small and large time steps.

Case 2: Riemann problem. The Riemann initial data is given as

$$f(0, x, v) = f_R \mathcal{H}(x) + f_L (1 - \mathcal{H}(x)), \quad f_R = 2.0, \quad f_L = 4.0 \quad (5.5)$$

with an inflow and outflow boundary condition. We set $A = 1$ and $\Omega_x = [-10, 10]$. The numerical solutions in the kinetic regime ($\varepsilon = 0.5$) and the diffusive regime ($\varepsilon = 10^{-6}$) are depicted in Figure 5.2. Two time steps are taken: $\Delta t = 0.5h$ and $\Delta t = 2.5h$. As we can see, our proposed schemes are stable for both time steps in both regimes. However, the larger the time step is, the bigger the deviation errors are, especially in the kinetic regime. The second-order scheme performs better than the first-order scheme.

5.1.3. One-group transport equation in a slab geometry

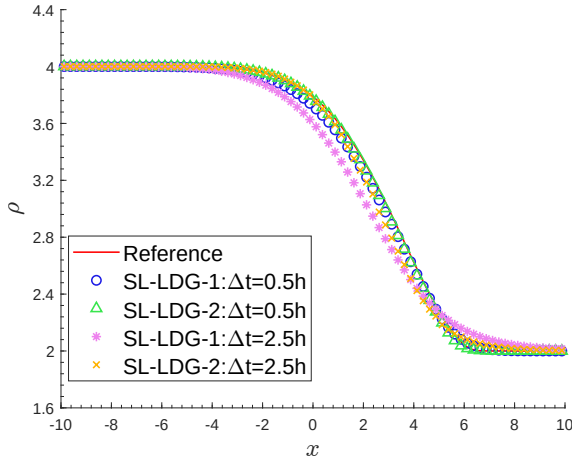
This is a transport problem with a continuous velocity space, and we take the discrete velocity method with a 16-point Gaussian-Legendre quadrature rule on $\Omega_v = [-1, 1]$. Here, we provide two typical numerical examples.

Case 1: Accuracy test. Let $\sigma_s = 1$, $\sigma_a = 0$ and $S \equiv 0$ and consider the following smooth initial data with a periodic boundary condition:

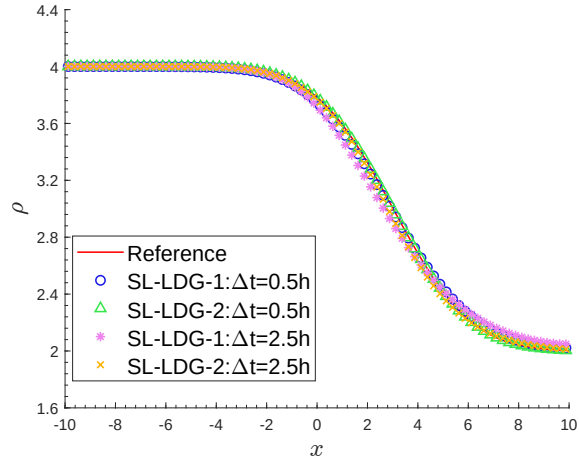
$$f(0, x, v) = 2 + \sin(x) - \varepsilon v \cos(x), \quad x \in [-\pi, \pi]. \quad (5.6)$$

Table 5.3: Accuracy test for advection-diffusion equation (Section 5.1.2). Numerical errors and convergences orders of the **SL-LDG- p** ($p = 1, 2$) methods in the diffusive regime $\varepsilon = 10^{-6}$.

Parameter		$\Delta t = 0.5\Delta x$				$\Delta t = 5.0\Delta x$			
p	N	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$
1	80	8.22E-02	-	3.14E-02	-	4.49E-01	-	1.29E-01	-
	160	4.19E-02	0.97	1.59E-02	0.99	2.57E-01	0.81	7.12E-02	0.86
	320	2.12E-02	0.99	7.93E-03	1.00	1.36E-01	0.91	3.76E-02	0.92
	640	1.06E-02	0.99	3.97E-03	1.00	7.02E-02	0.95	1.93E-02	0.96
2	80	2.57E-03	-	8.81E-04	-	2.96E-01	-	7.42E-02	-
	160	6.35E-04	2.02	2.20E-04	2.00	6.71E-02	2.14	1.68E-02	2.14
	320	1.56E-04	2.03	5.46E-05	2.01	1.64E-02	2.03	4.11E-03	2.03
	640	3.92E-05	1.99	1.37E-05	2.00	3.99E-03	2.04	1.00E-03	2.04



(a) kinetic regime: $\varepsilon = 0.5$, $T = 3.0$



(b) diffusive regime: $\varepsilon = 10^{-6}$, $T = 3.0$

Figure 5.2: Riemann problem for advection-diffusion equation (Section 5.1.2). Density ρ of the **SL-LDG- p** ($p = 1, 2$) solutions at different regimes. $h = 0.25$. The reference solution is computed by the first-order finite difference method based on micro-macro decomposition with $h = 0.04$ and $\Delta t = 0.01$.

437 The convergence study is carried out by taking $T = 1$, $N = 40 \cdot 2^k$ and $\Delta t = T/(3 \cdot 2^k) \approx 2.12\Delta x$.
438 Since the exact solutions of f and ρ are not available, we compute the errors by comparing two
439 solutions on two consecutive meshes. Specifically, we define

$$RE_{N,r}^\rho = \|\rho_h(T, x) - \rho_{h/2}(T, x)\|_{L^r}, \quad RO_{N,r}^\rho = \log_2(RE_{N,r}^\rho / RE_{2N,r}^\rho), \quad (5.7)$$

440 for $r = 1, \infty$. Table 5.4 validates the convergence order of both first- and second-order schemes
441 from kinetic regime $\varepsilon = 0.5$ to diffusive regime $\varepsilon = 10^{-6}$.

442 **Case 2: Boundary condition test.** Given $\sigma_s = 1, \sigma_a = 0$ and $S \equiv 0$, we consider
443 a benchmark problem from [12, 10] with an isotropic Dirichlet boundary condition and zero
444 initial data on $\Omega_x = [0, 1]$, namely,

$$f_L(t, v) = 1, \quad f_R(t, v) = 0, \quad f(0, x, v) = 0, \quad x \in \Omega_x. \quad (5.8)$$

445 The numerical results shown in Figure 5.3 demonstrate the robustness of our proposed schemes,

Table 5.4: Accuracy test for one-group transport equation (Section 5.1.3). Numerical errors and convergence orders of the SL-LDG- p ($p = 1, 2$) methods in L^1 and L^∞ norms. $T = 1$.

Parameters		$p = 1$				$p = 2$			
ε	N	$RE_{N,1}^\rho$	$RO_{N,1}^\rho$	$RE_{N,\infty}^\rho$	$RO_{N,\infty}^\rho$	$RE_{N,1}^\rho$	$RO_{N,1}^\rho$	$RE_{N,\infty}^\rho$	$RO_{N,\infty}^\rho$
0.5	80	1.32E-01	-	3.29E-02	-	2.23E-03	-	1.10E-03	-
	160	6.99E-02	0.92	1.75E-02	0.91	5.58E-04	2.00	2.69E-04	2.03
	320	3.62E-02	0.95	9.05E-03	0.95	1.40E-04	2.00	6.65E-05	2.02
	640	1.84E-02	0.97	4.61E-03	0.97	3.49E-05	2.00	1.66E-05	2.01
10^{-2}	80	1.31E-01	-	3.27E-02	-	3.44E-03	-	2.23E-03	-
	160	6.15E-02	1.09	1.54E-02	1.09	1.02E-03	1.76	6.63E-04	1.75
	320	3.00E-02	1.04	7.50E-03	1.03	2.53E-04	2.01	1.65E-04	2.00
	640	1.49E-02	1.01	3.72E-03	1.01	6.21E-05	2.02	4.08E-05	2.02
10^{-6}	80	1.41E-01	-	3.51E-02	-	4.09E-03	-	2.65E-03	-
	160	7.03E-02	1.00	1.76E-02	1.00	1.02E-03	2.00	6.64E-04	2.00
	320	3.51E-02	1.00	8.78E-03	1.00	2.55E-04	2.00	1.66E-04	2.00
	640	1.76E-02	1.00	4.39E-03	1.00	6.37E-05	2.00	4.15E-05	2.00

which are AP when it is close to the diffusive limit, have large time step stability, and are effective for such a Dirichlet boundary condition.

5.2. Multi-dimensional transport problems

Now we consider some multi-dimensional transport problems (2.5) to demonstrate the efficiency of our proposed schemes. For high-dimensional problems, the velocity is defined as vectors on a unit sphere $\Omega_{\mathbf{v}}$, and the average in the velocity space is defined as $\langle \cdot \rangle = \frac{1}{4\pi} \int_{\Omega_{\mathbf{v}}} \cdot d\mathbf{v}$. For velocity discretization, we adopt the Lebedev quadrature rule with $N_{\mathbf{v}}$ points [50]. Periodic or zero boundary conditions are considered.

By setting σ_s , σ_a , $\sigma = \sigma_s + \varepsilon^2 \sigma_a$ and following the procedure in Section 2, we can obtain a similar approximation model, which is [27]

$$\partial_t \rho + e^{-\sigma(t-t^n)/\varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} g(t^n, \mathbf{x}^n, \mathbf{v}) \rangle = \nabla_{\mathbf{x}} \cdot \left(\frac{1}{\sigma} (1 - e^{-\sigma(t-t^n)/\varepsilon^2}) \nabla_{\mathbf{x}} \rho \right) - \sigma_a \rho + S, \quad (5.9a)$$

$$\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{\sigma_s}{\varepsilon^2} (\rho - f) - \sigma_a f + S. \quad (5.9b)$$

5.2.1. The reduced 2D problem

Firstly, we carry out an accuracy test with the exact solution from [51]:

$$f(t, x, y, v_x, v_y) = e^{-t} \sin^2(2\pi x) \sin^2(2\pi y) \left(1 + \varepsilon \left(\frac{v_y + v_y^3}{3} \right) \right), \quad (x, y) \in [0, 1]^2. \quad (5.10)$$

Let $\sigma_s = 1$, $\sigma_a = 0$, then the source term S can be obtained via

$$S(t, x, y, v_x, v_y) = \partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \frac{1}{\varepsilon^2} (f - \langle f \rangle). \quad (5.11)$$

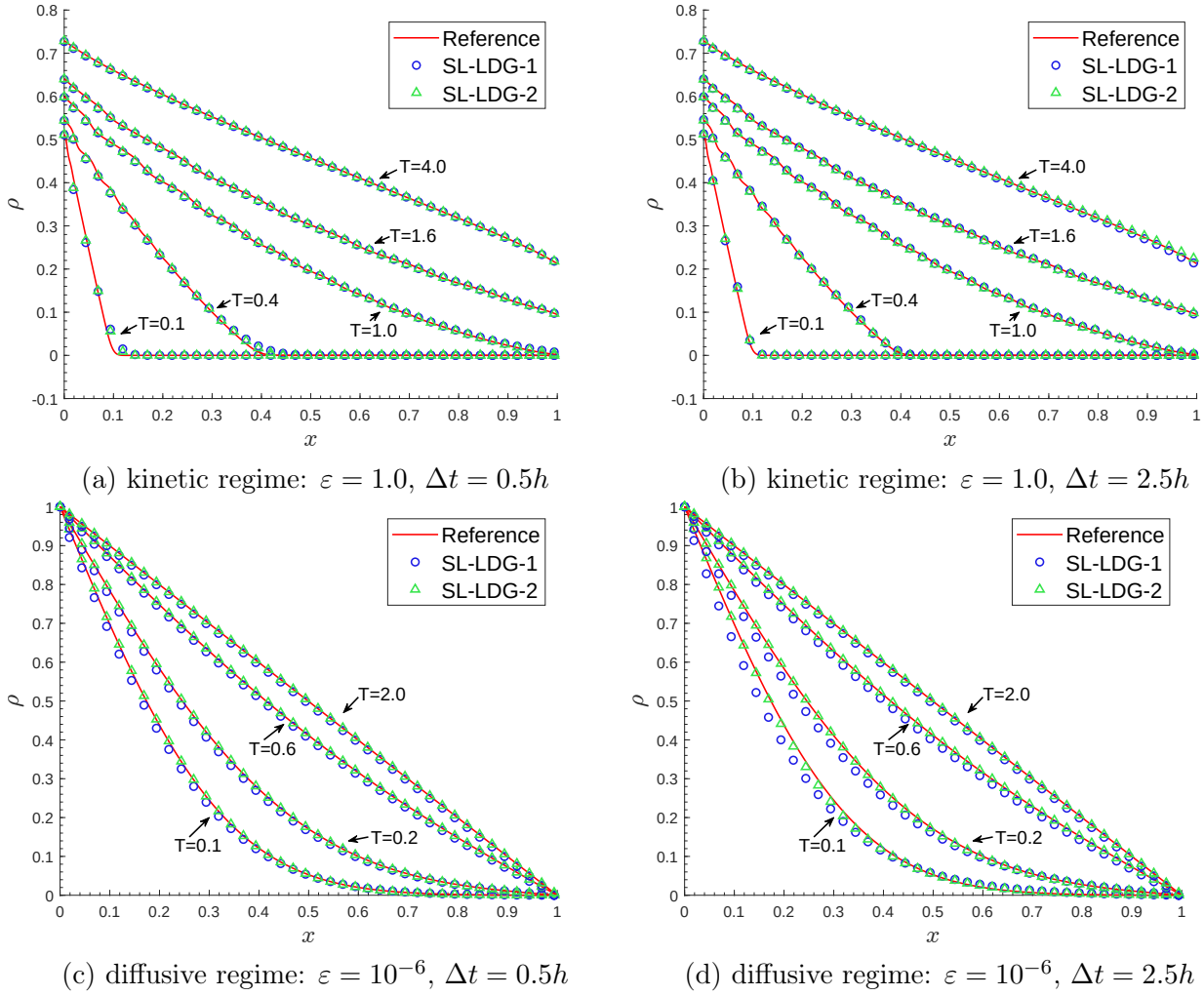


Figure 5.3: Isotropic Dirichlet boundary test for one-group transport equation (Section 5.1.3). Density ρ of the SL-LDG- p ($p = 1, 2$) solutions at different regimes. $h = 0.0125$. The reference solution is computed by the first-order finite difference method based on micro-macro decomposition with $h = 0.001$ and $\Delta t = 0.00025$.

As the source term S depends on time t and velocity $\mathbf{v}$, we derive an approximation model (5.12) instead of (5.9) via the same reformulation approach in Section 2, which is

$$\begin{aligned} \partial_t \rho + e^{-\sigma(t-t^n)/\varepsilon^2} \nabla_{\mathbf{x}} \cdot \langle \mathbf{v} g(t^n, \mathbf{x}^n, \mathbf{v}) \rangle = & \nabla_{\mathbf{x}} \cdot \left(\frac{1}{\sigma} (1 - e^{-\sigma(t-t^n)/\varepsilon^2}) \nabla_{\mathbf{x}} \rho \right) - \sigma_a \rho + \langle S \rangle \\ & - \varepsilon \nabla_{\mathbf{x}} \cdot \left(\frac{1}{\sigma} (1 - e^{-\sigma(t-t^n)/\varepsilon^2}) \langle \mathbf{v} S \rangle \right), \end{aligned} \quad (5.12a)$$

$$\partial_t f + \frac{1}{\varepsilon} \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{\sigma_s}{\varepsilon^2} (\rho - f) - \sigma_a f + S. \quad (5.12b)$$

For this test, we take $T = 1.0$, $N_{\mathbf{v}} = 590$, $\varepsilon = 1, 10^{-2}, 10^{-6}$ and $\Delta t = 2.5h$. The results are shown in Table 5.5, which validate the desired convergence order of the proposed methods in both kinetic and diffusive regimes. Similarly, order reductions are observed in the intermediate regime due to the modeling approximation errors.

Secondly, we consider a Gaussian initial data centered at the origin with variance ζ^2 [51, 27]

Table 5.5: 2D Accuracy test (Section 5.2.1). Numerical errors and convergence orders of the SL-LDG- p ($p = 1, 2$) methods in L^1 and L^∞ norms. $T = 1$.

Parameter		$p = 1$				$p = 2$			
ε	$N_x \times N_y$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$
1.0	16×16	2.68E-02	-	1.20E-01	-	1.26E-02	-	4.98E-02	-
	32×32	1.42E-02	0.92	6.20E-02	0.95	3.21E-03	1.97	1.36E-02	1.88
	64×64	7.15E-03	0.99	3.18E-02	0.96	8.36E-04	1.94	3.59E-03	1.92
10^{-2}	16×16	2.17E-02	-	1.07E-01	-	5.20E-03	-	3.21E-02	-
	32×32	1.09E-02	1.00	5.40E-02	0.99	2.20E-03	1.24	7.22E-03	2.15
	64×64	5.59E-03	0.96	2.60E-02	1.06	1.43E-03	0.62	4.13E-03	0.81
10^{-6}	16×16	2.17E-02	-	1.07E-01	-	4.68E-03	-	3.69E-02	-
	32×32	1.09E-02	1.00	5.42E-02	0.98	1.18E-03	1.98	9.90E-03	1.90
	64×64	5.48E-03	0.99	2.67E-02	1.02	2.97E-04	1.99	2.52E-03	1.97

466

$$f(t=0, x, y, v_x, v_y) = \frac{1}{4\pi\zeta^2} \exp\left(-\frac{x^2 + y^2}{4\zeta^2}\right), \quad (5.13)$$

467 and set $\sigma_a = 0$, $S \equiv 0$. We are interested in two challenging cases as follows:

468 **Case 1: Spatially varying scattering coefficient σ_s in the intermediate regime.**

469 Let $\varepsilon = 0.01$, $\sigma_a = S = 0$, $\zeta^2 = 10^{-2}$ and σ_s varies in space as

$$\sigma_s(x, y) = \begin{cases} 0.999c^4(c + \sqrt{2})^2(c - \sqrt{2})^2 + 0.001, & c = \sqrt{x^2 + y^2} < 1; \\ 1, & \text{otherwise.} \end{cases} \quad (5.14)$$

470 We take $N_x = N_y = 128$, $N_v = 590$. This is a challenging problem because the collision
471 coefficient σ_s/ε varies in a large range $[0.1, 100]$. For existing methods [22, 23, 45], they require
472 a time step $\Delta t = 0.1 \min(\sigma_s)h^2 + 0.1\varepsilon h$, which is subject to a small time step in the intermediate
473 regime. But as shown in Figure 5.4, our methods are still stable with a ten times or a forty
474 times larger time step. At the initial time layer, the first- and second-order schemes show no
475 big difference. However, with such a large time step, the first-order SL-LDG-1 scheme has
476 large deviation errors again. The second-order SL-LDG-2 scheme is better than the first-order
477 scheme and is more accurate, even with large time steps.

478 **Case 2: Line source problem.** The line source problem describes an isotropic radiation
479 impulse with total energy concentrated on the z -axis [53, 54, 55, 56, 51]. Hence, the initial data
480 involves a Dirac delta function. For numerical simulations, we approximate the initial data by
481 a steep Gaussian distribution (5.13) with $\zeta^2 = 3.2 \times 10^{-4}$. Besides, we set $\sigma_s = 1$, $\sigma_a = 0$, $S \equiv 0$
482 and $\Omega_{\mathbf{x}} = [-0.6, 0.6]^2$. We run the test with $\varepsilon = 1$, $T = 0.5$, $N_x = N_y = 128$ and $N_v = 5810$.

483 This example is in the kinetic regime and requires a large number of discrete ordinates
484 to avoid a Ray effect [53, 51]. This example shows the advantages of employing high order
485 methods. As depicted in Figure 5.2.1, the first-order method appears to converge not very
486 well, and we can still observe some Ray effects, while the second-order method fits the physical
487 solution much better under a large time step $\Delta t = 2.5h$.

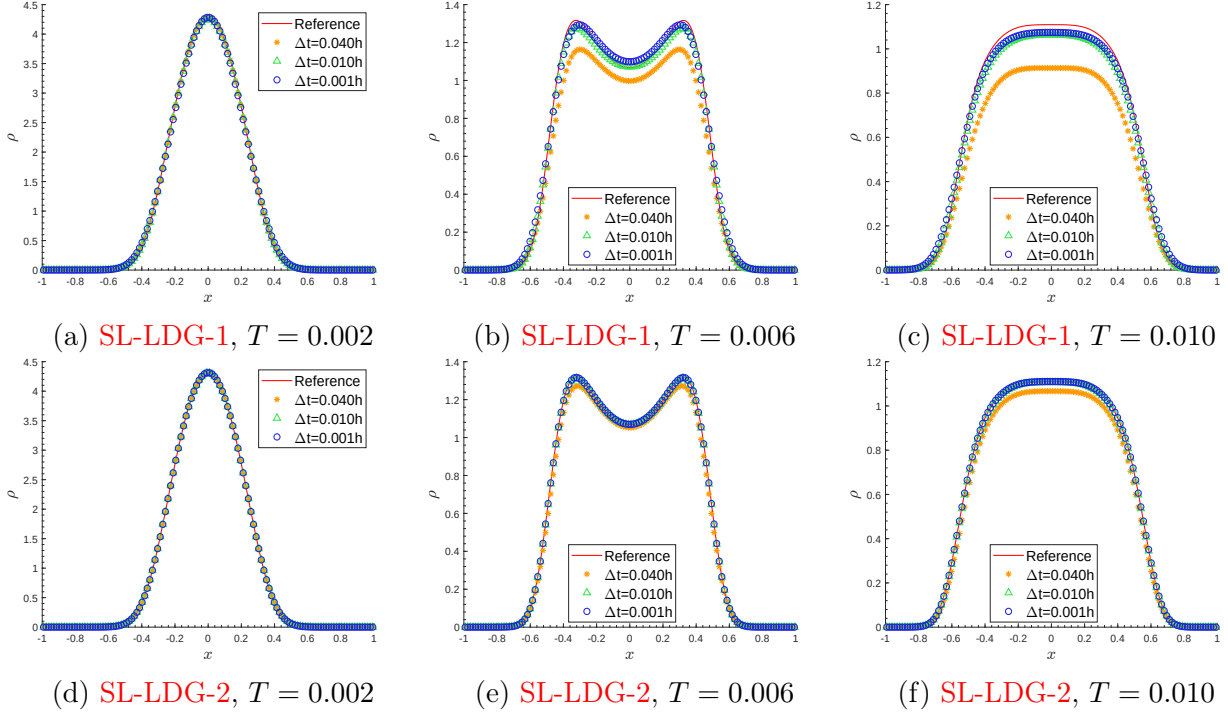


Figure 5.4: Varying scattering problem for 2D transport problem (Section 5.2.1). Density ρ at $y = 0$ of the SL-LDG- p ($p = 1, 2$) solutions at $T = 0.002, 0.006, 0.010$ with $N_x = N_y = 128$, $N_v = 590$ and different time steps Δt . The reference solutions are computed by the second-order finite difference method on staggered grids [52] with $h = 0.005$ and $\Delta t = 10^{-5}$.

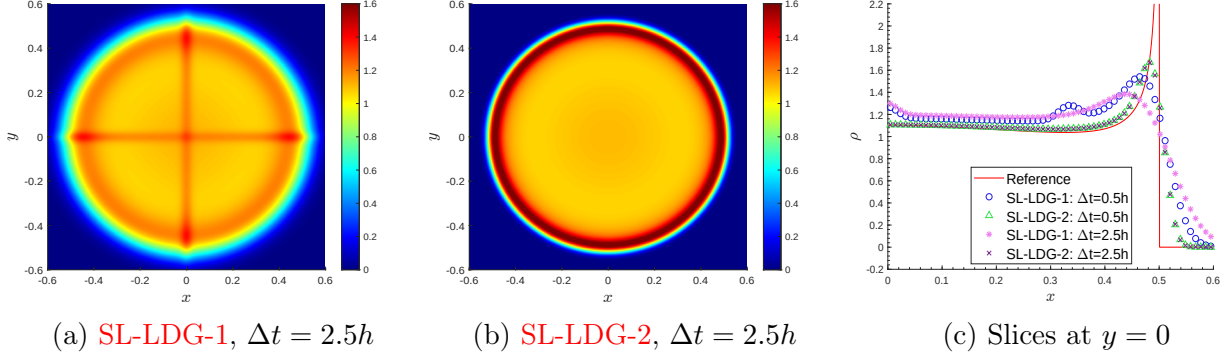


Figure 5.5: Line source test for 2D transport problem (Section 5.2.1). Left and Middle: Density ρ of the SL-LDG- p ($p = 1, 2$) solutions at $T = 0.5$. Right: Slice at $y = 0$ with a reference solution from [57, 58]. Parameters are $\varepsilon = 1$, $N_x = N_y = 128$, $N_v = 5810$. $T = 0.5$.

5.2.2. 3D problem

Case 1: Accuracy test. Consider $\Omega_x = [-1, 1]^3$, $\sigma_s = 1$, $\sigma_a = 0$ and exact solution

$$\rho(t, x, y, z) = e^{-t} \sin(\pi x) \sin(\pi y) \sin(\pi z), \quad g = -(\mathbf{v} \cdot \nabla_x \rho), \quad f = \rho + \varepsilon g. \quad (5.15)$$

The source term S is computed again by (5.12). We simulate with $N_v = 170$, $\varepsilon = 1, 10^{-2}, 10^{-6}$, $T = 1.5$, $\Delta t = 2.5h$. In Table 5.6, the convergence orders for first-order and second-order schemes with different ε 's are presented. The desired orders of accuracy with a large time step can be observed.

Table 5.6: 3D accuracy test (Section 5.2.2). Numerical errors and convergence orders of the **SL-LDG- p** ($p = 1, 2$) methods in L^1 and L^∞ norms. $T = 1.5$.

Parameter		$p = 1$				$p = 2$			
ε	$N_x \times N_y \times N_z$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$	$E_{N,1}^\rho$	$O_{N,1}^\rho$	$E_{N,\infty}^\rho$	$O_{N,\infty}^\rho$
1	$8 \times 8 \times 8$	1.93E-02	-	9.03E-01	-	3.24E-03	-	1.81E-01	-
	$16 \times 16 \times 16$	1.08E-02	0.83	5.98E-01	0.60	4.76E-04	2.77	3.00E-02	2.59
	$32 \times 32 \times 32$	5.55E-03	0.97	3.10E-01	0.95	1.09E-04	2.13	5.38E-03	2.48
10^{-2}	$8 \times 8 \times 8$	4.08E-03	-	1.87E-01	-	1.27E-03	-	9.70E-02	-
	$16 \times 16 \times 16$	1.99E-03	1.03	8.93E-02	1.06	3.23E-04	1.98	2.71E-02	1.84
	$32 \times 32 \times 32$	9.90E-04	1.01	4.44E-02	1.01	8.64E-05	1.90	7.41E-03	1.87
10^{-6}	$8 \times 8 \times 8$	4.08E-03	-	1.86E-01	-	1.28E-03	-	9.75E-02	-
	$16 \times 16 \times 16$	1.99E-03	1.03	8.91E-02	1.06	3.28E-04	1.96	2.75E-02	1.82
	$32 \times 32 \times 32$	9.90E-04	1.01	4.44E-02	1.00	8.32E-05	1.98	7.16E-03	1.94

Case 2: Point source problem. Set $\varepsilon = 1$, $\Omega_{\mathbf{x}} = [-0.6, 0.6]^3$, $\sigma_s = 1$, $\sigma_a = 0$, $S \equiv 0$ and consider the following initial data:

$$f(t = 0, x, y, z, v_x, v_y, v_z) = \frac{1}{4\pi\zeta^2} \exp\left(-\frac{x^2 + y^2 + z^2}{4\zeta^2}\right), \quad \zeta^2 = 3.2 \times 10^{-4}. \quad (5.16)$$

Here we compare the numerical behavior of the **SL-LDG-2** method with that of the StaRMAP method [52]. The StaRMAP method solves the spherical harmonics moment (P_N) systems using a second-order staggered grid finite difference method [59]. In the simulation, we take the StaRMAP method by setting $N = 600$ in P_N , while $N = N_v = 1202$ in S_N for our **SL-LDG-2** method with comparable degrees of freedom. We take the mesh size $h = 0.024$ and time step $\Delta t = 1/128$. As depicted in Figure 5.6, the StaRMAP solution exhibits notable numerical oscillations due to the wave effect of the P_N method. Our method is almost oscillation-free, but it exhibits a little ray effect due to the S_N method.

Secondly, we evaluate the parallel efficiency of the proposed methods with an OpenMP API. We refer to an R-step for solving the macroscopic equation and an F-step for solving the microscopic equation. In the R-step, the transport term involves a numerical integration in velocity, which can be computed in a parallel way. In the F-step, the distribution function with different velocities can be updated simultaneously since the density is known. As illustrated in Table 5.7, the speedup trend for the F-step parallelization is nearly linear. Surprisingly, the R-step parallelization enjoys superlinear speedup. This remarkable performance can be attributed to the efficient cache exploitation on a shared memory architecture [60, 61]. As a result, we have significantly reduced the total runtime from 13552.66 seconds (225.87 minutes) to 1047.95 seconds (17.47 minutes) via the multi-threading technique (single thread to 16 threads). In other words, the computational speed increased by more than 10 times.

Table 5.7: 3D Point source problem (Section 5.2.2). Parallel efficiency of the SL-LDG-2 method. Parameters are $N_x = N_y = N_z = 50$, $N_v = 1202$, $T = 0.5$ and $\Delta t = 0.025$. P is the number of threads; RT is the running time; S is the speedup ratio; E is the parallel efficiency. The time unit is second.

P	RT_{total}	S_{total}	E_{total}	RT_ρ	S_ρ	E_ρ	RT_f	S_f	E_f
1	13552.66	-	-	8364.99	-	-	4163.34	-	-
2	6835.17	1.98	99.14%	3947.65	2.12	105.95%	2297.08	1.81	90.62%
4	3512.93	3.86	96.45%	1931.29	4.33	108.28%	1247.16	3.34	83.46%
8	1641.28	8.26	103.22%	764.02	10.95	136.86%	647.89	6.43	80.32%
16	1047.95	12.93	80.83%	466.11	17.95	112.17%	355.25	11.72	73.25%

Remark 5.1. Let us discuss the memory usage of the proposed methods. A floating-point number is referred as a unit. Our analysis is confined to a uniform rectangular mesh. We denote d as the number of spatial dimensions, N_x as the count of elements along each spatial dimension and N_{dof} as the number of degrees of freedom (DoFs) per element in a DG method. In particular, $N_{dof} = \frac{(k+d)!}{d!}$ for a $\mathcal{P}^k$ space. We recall that N_v is the number of discrete velocities. Firstly, the DoFs associated with f and ρ consume $O(N_v N_x^d N_{dof})$ units storage. Secondly, diffusion operators $l_K(\cdot, \cdot)$ and $d_K(\cdot, \cdot)$ in (3.10) need $O(N_x^d N_{dof}^2)$ units for sparse matrices storage. Thirdly, the action of operators $b_K(\cdot, \cdot)$ in (3.10) and $a_K(\cdot, \cdot)$ in (3.21) on the DoFs of f and ρ is equivalent to matrix-vector products. The underlying matrices involve integrals of type (c) and (d) in Remark 3.1, which are dependent on the time step size. As a uniform time step is used, store these integrals at once can significantly save computational cost. For a uniform rectangular mesh, integral evaluation only need to be done within a hypercube $[0, 1]^d$ for each discrete velocity. With a fixed velocity, through the clipping operation, there are at most $N_e = 2^d$ sub-elements and $N_f = d2^d$ sub-faces generated. The operator $a_K(\cdot, \cdot)$ stores 1 matrix of volume integrals at each sub-element. The operator $b_K(\cdot, \cdot)$ stores d matrices of volume integrals at each sub-element and 1 matrix of flux integrals at each sub-face. Each matrix is of shape $N_{dof} \times N_{dof}$. As a result, the total storage for operators $b_K(\cdot, \cdot)$ and $a_K(\cdot, \cdot)$ is no more than $((d+1)N_e + N_f)N_v N_{dof}^2 = (2d+1)2^d N_v N_{dof}^2$ units. In summary, with a uniform mesh, the total memory usage of proposed SL-LDG- p - q methods is almost at the same level as that of a standard implicit upwind DG scheme.

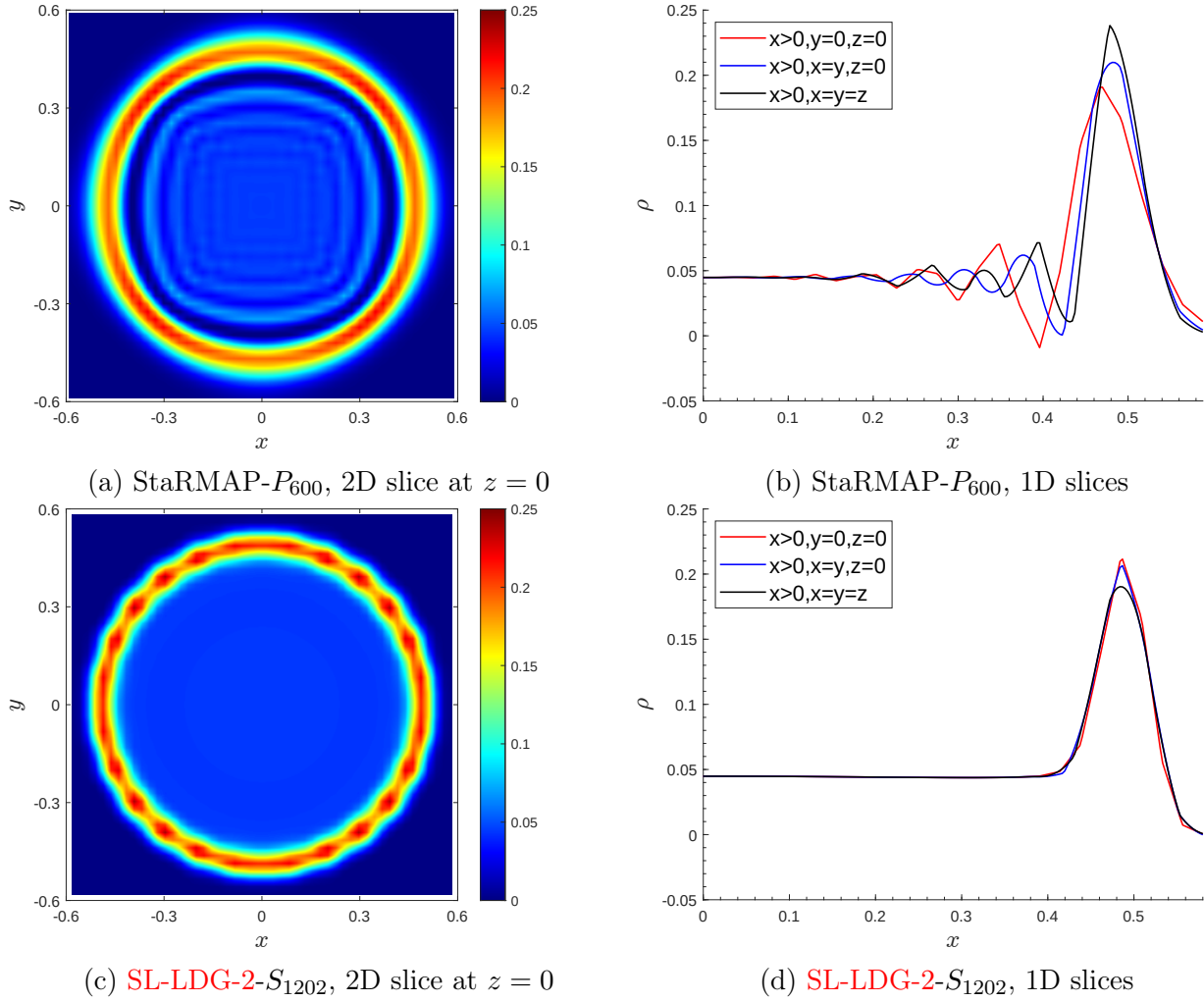


Figure 5.6: 3D Point source problem(Section 5.2.2). Comparison of the StaRMAP and **SL-LDG-2** solutions at $T = 0.5$. (a)(b): Density plots of the StaRMAP solution for a P_{600} system. (c)(d): Density plots of the **SL-LDG-2** solution for a S_{1202} system ($N_v = 1202$). Both tests are run with $N_x = N_y = N_z = 50$ and $\Delta t = T/64$.

Case 3: Varying scattering problem in the intermediate regime. Finally we consider a 3D extension of Case 1 in Section 5.2.1. With such an example, we can clearly see the superiority of our methods for solving such problems in the intermediate regime. Set $\Omega_x = [-1, 1]^3$, $\sigma_a = S = 0$, $\varepsilon = 0.01$ and

$$\sigma_s(x, y, z) = \begin{cases} 0.999c^4(c + \sqrt{2})^2(c - \sqrt{2})^2 + 0.001, & c = \sqrt{x^2 + y^2 + z^2} < 1; \\ 1, & \text{otherwise.} \end{cases} \quad (5.17)$$

The initial condition is still (5.16) but with $\zeta^2 = 10^{-2}$. We also compare the proposed **SL-LDG-2** method with the StaRMAP method. The StaRMAP method is only available for $\varepsilon = 1$. To make a comparison, we scaled the termination time and scattering coefficient by taking $T^{StaRMAP} = T/\varepsilon$ and $\sigma_s^{StaRMAP} = \sigma_s/\varepsilon^2$, where T, σ_s are used for the **SL-LDG-2** method. Besides, we solve a P_{13} system for the StaRMAP method and a S_{194} system for the **SL-LDG-2** method, with comparable degrees of freedom in the velocity space. The numerical results are presented in Figure 5.7. Here we fix the final time T but take different time steps to arrive at

the final time, corresponding to different time step sizes. The two methods perform similarly with small time steps. However, with a larger time step, the StaRMAP method would blow up after some time, while the SL-LDG-2 method remains stable and match the reference solution well.

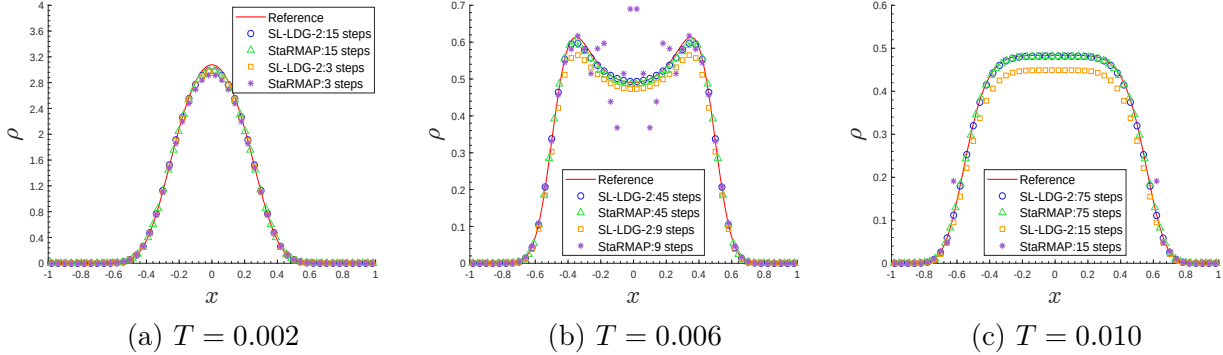


Figure 5.7: Varying scattering problem for 3D transport problem (Section 5.2.2). Comparison of the SL-LDG-2 method and StaRMAP method under different time steps. Density profile ρ at $y = z = 0$ is plotted at $T = 0.002, 0.006, 0.010$. Parameters are $\varepsilon = 0.01, N_x = N_y = N_z = 50$. The StaRMAP solutions are for P_{13} system while SL-LDG-2 solutions are for S_{194} system ($N_v = 194$). The reference solutions are computed by a first-order finite difference method on staggered grids with $N_x = N_y = N_z = 150$ and $\Delta t = 0.1\varepsilon h + 0.1h^2$.

6. Conclusions and outlook

In this paper, we have developed a class of uniformly unconditionally stable AP SL DG methods for solving kinetic transport equations under a diffusive scaling. We design both first-order and second-order SL DG methods for an approximation model, leveraging the strength of SL techniques in a weak DG formulation to enhance computational efficiency. Furthermore, within the DG framework, it is possible to utilize an oscillation-free damping limiter in tandem with an implicit time discretization, surpassing the performance of conventional slope limiters. Numerical experiments spanning from 1D to 3D problems have provided empirical validation of the expected orders of accuracy, robustness, and high parallel efficiency inherent in our methods. The effectiveness and efficiency of our proposed approaches are particularly pronounced when addressing high-dimensional multiscale problems.

In the future, it would be intriguing to explore more general collision operators, such as those associated with frequency-dependent kinetic problems. Additionally, we plan to investigate uniformly unconditional stability through energy stability analyses.

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720 Appendix A. Proof of Lemma 2.1

721 *Proof.* Let $E(s) := \int_0^s e^{-\alpha(s-\tau)} (\eta(\tau) - \eta(s)) d\tau$, and by using the Taylor series expansion we
 722 have

$$\begin{aligned} E(s) &= \int_0^s e^{-\alpha(s-\tau)} \left[\sum_{i=1}^{\infty} \frac{(-1)^i (s-\tau)^i}{i!} \eta^{(i)}(s) d\tau \right] \\ &= \sum_{i=1}^{\infty} \frac{(-1)^i}{i!} \eta^{(i)}(s) \int_0^s e^{-\alpha(s-\tau)} (s-\tau)^i d\tau \\ &= \sum_{i=1}^{\infty} \frac{(-1)^i}{\beta^{i+1} i!} \beta^{i+1} \eta^{(i)}(s) \int_0^s e^{-\alpha(s-\tau)} (s-\tau)^i d\tau. \end{aligned} \quad (\text{A.1})$$

723 The parameter β^{i+1} is added to match the scale of $\eta^{(i)}(s)$. As we can see, we will take $\beta = \varepsilon$ in
 724 (B.1) in the proof of Theorem 2.2. We denote $A_i = \int_0^s e^{-\alpha(s-\tau)} (s-\tau)^i d\tau$ and $B_i = s^i e^{-\alpha s}$, and
 725 by mathematical induction we obtain

$$B_0 = e^{-\alpha s}, \quad A_0 = \alpha^{-1}(1 - B_0), \quad A_i = \alpha^{-1}(iA_{i-1} - B_i), \quad i \geq 1. \quad (\text{A.2})$$

726 This relation also yields

$$A_i = \alpha^{-1} \left(\alpha^{-i}(i!) - \sum_{j=0}^i \frac{i!}{(i-j)!} \alpha^{-j} B_{i-j} \right) = \alpha^{-(i+1)}(i!) \left(1 - \sum_{j=0}^i \frac{\alpha^j}{j!} B_j \right). \quad (\text{A.3})$$

727 Then, we have

$$E(s) = \sum_{i=1}^{\infty} \beta^{-(i+1)} C_i D_i, \quad C_i = (-1)^i \alpha^{-(i+1)} \beta^{i+1} \eta^{(i)}(s), \quad D_i = 1 - \sum_{j=0}^i \frac{\alpha^j}{j!} B_j, \quad (\text{A.4})$$

728 and by the assumption $|\beta^{i+1} \eta^{(i)}(s)| \leq M$ for all $i \geq 1$,

$$|C_i| \leq \alpha^{-(i+1)} M, \quad |D_i| = e^{-\alpha s} \left(e^{\alpha s} - \sum_{j=0}^i \frac{(\alpha s)^j}{j!} \right) = e^{-\alpha s} \sum_{j=i+1}^{\infty} \frac{(\alpha s)^j}{j!}, \quad 0 < D_i < 1. \quad (\text{A.5})$$

729 With these, we arrive at

$$\begin{aligned} |E(s)| &\leq \sum_{i=1}^{\infty} |C_i| |D_i| \leq M e^{-\alpha s} \sum_{i=1}^{\infty} \sum_{j=i+1}^{\infty} (\alpha \beta)^{-(i+1)} \frac{(\alpha s)^j}{j!} \\ &= M e^{-\alpha s} \sum_{i=2}^{\infty} \sum_{j=i}^{\infty} \frac{(\alpha \beta)^{-i} (\alpha s)^j}{j!} \\ &= M e^{-\alpha s} \sum_{j=2}^{\infty} \sum_{i=2}^j \frac{(\alpha \beta)^{-i} (\alpha s)^j}{j!} =: MU(\alpha, \beta, s). \end{aligned} \quad (\text{A.6})$$

730 Let $u = (\alpha\beta)^{-1}$ and $v = \alpha s$, we rewrite the upper bound U as follows:

$$U = e^{-\alpha s} \sum_{j=2}^{\infty} \sum_{i=2}^j \frac{(\alpha\beta)^{-i} (\alpha s)^j}{j!} = e^{-v} \sum_{j=2}^{\infty} \sum_{i=2}^j \frac{u^i v^j}{j!} = e^{-v} \sum_{j=2}^{\infty} \frac{w_j v^j}{j!} \quad (\text{A.7})$$

731 where

$$w_j = \begin{cases} \frac{u^2(1-u^{j-1})}{1-u}, & u \neq 1; \\ j-1, & u = 1. \end{cases} \quad (\text{A.8})$$

732 When $u = 1$, we have

$$\begin{aligned} U &= e^{-v} \sum_{j=2}^{\infty} \frac{(j-1)v^j}{j!} = v^2 e^{-v} \sum_{j=2}^{\infty} \frac{d}{dv} \left(\frac{v^{j-1}}{j!} \right) = v^2 e^{-v} \frac{d}{dv} \left(v^{-1} \sum_{j=2}^{\infty} \frac{v^j}{j!} \right) \\ &= v^2 e^{-v} \frac{d}{dv} (v^{-1}(e^v - 1 - v)) = e^{-v} - 1 + v = e^{-\alpha s} - 1 + \alpha s. \end{aligned} \quad (\text{A.9})$$

733 When $u \neq 1$, we get

$$\begin{aligned} U &= \frac{u^2 e^{-v}}{1-u} \sum_{j=2}^{\infty} \frac{(1-u^{j-1})v^j}{j!} = \frac{u^2 e^{-v}}{1-u} \left[\sum_{j=2}^{\infty} \frac{v^j}{j!} - u^{-1} \sum_{j=2}^{\infty} \frac{(uv)^j}{j!} \right] \\ &= \frac{u^2 e^{-v}}{1-u} [(e^v - 1 - v) - u^{-1}(e^{uv} - 1 - uv)] = \frac{u^2 e^{-v}}{1-u} [e^v - 1 - u^{-1}e^{uv} + u^{-1}] \\ &= \frac{u^2(1-e^{-v})}{1-u} + \frac{ue^{-v}(1-e^{uv})}{1-u} = \frac{(\alpha\beta)^{-2}(1-e^{-\alpha s})}{1-(\alpha\beta)^{-1}} + \frac{(\alpha\beta)^{-1}(e^{-\alpha s} - e^{-(\alpha\beta^{-1})s})}{1-(\alpha\beta)^{-1}}. \end{aligned} \quad (\text{A.10})$$

734 □

735 Appendix B. Proof of Theorem 2.2

736 *Proof.* Let $\tau = t - t_0$, $s = t_1 - t_0 = \Delta t$, $\eta(\tau) = \frac{d\rho}{d\tau}(\tau)$, $\alpha = \varepsilon^{-2}$, $\beta = \varepsilon$. The assumption
737 $|\beta^{i+1}\eta^{(i)}(s)| \leq M$ now becomes:

$$|\beta^{k+1}\eta^{(k)}(s)| = |(\varepsilon\partial_t + \mathbf{v} \cdot \nabla_{\mathbf{x}})^{k+1}\rho(s)| \leq M, \quad \forall k \geq 1. \quad (\text{B.1})$$

738 By Lemma 2.1, we have

$$\begin{aligned} \|\mathbf{F} - \mathbf{G}\|_{\infty} &\leq \left| \left\langle \frac{\|\mathbf{v}\|_{\infty}}{\varepsilon} \int_{t_0}^{t_1} e^{-(t_1-t)/\varepsilon^2} \left(\frac{d\rho}{dt}(t) - \frac{d\rho}{dt}(t_1) \right) dt \right\rangle \right| \\ &= \left| \left\langle \frac{\|\mathbf{v}\|_{\infty}}{\varepsilon} \int_0^s e^{-(s-\tau)/\varepsilon^2} (\eta(\tau) - \eta(s)) d\tau \right\rangle \right| \\ &\leq \left\langle \frac{\|\mathbf{v}\|_{\infty}}{\varepsilon} \left| \int_0^s e^{-(s-\tau)/\varepsilon^2} (\eta(\tau) - \eta(s)) d\tau \right| \right\rangle \\ &\leq \frac{\langle \|\mathbf{v}\|_{\infty} \rangle}{\varepsilon} MU(\varepsilon, \Delta t) = CW(\varepsilon, \Delta t) \end{aligned} \quad (\text{B.2})$$

739 where $\|\mathbf{x}\|_{\infty} = \max_i |x_i|$, $C = \langle \|\mathbf{v}\|_{\infty} \rangle M > 0$ is a constant independent of ε and Δt ; $W(\varepsilon, \Delta t)$
740 is given as follows:

$$W(\varepsilon, \Delta t) = \frac{1}{\varepsilon} U(\varepsilon, \Delta t) = \begin{cases} e^{-\Delta t} - 1 + \Delta t, & \varepsilon = 1; \\ \frac{\varepsilon(1-e^{-\Delta t/\varepsilon^2})}{1-\varepsilon} - \frac{e^{-(1-\varepsilon)\Delta t/\varepsilon^2} - e^{-\Delta t/\varepsilon^2}}{1-\varepsilon}, & \varepsilon \neq 1. \end{cases} \quad (\text{B.3})$$

741 (1) For $\varepsilon = 1$, from a Taylor series expansion of $e^{-\Delta t}$, (B.3) implies that $\|\mathbf{F} - \mathbf{G}\|_\infty = O(\Delta t^2)$
 742 as $\Delta t \rightarrow 0$. (2) For $0 < \varepsilon < 1$, on the one hand, we have $W(\varepsilon, \Delta t) < \frac{\varepsilon}{1-\varepsilon}$ as $e^{-(1-\varepsilon)\Delta t/\varepsilon^2} >$
 743 $e^{-\Delta t/\varepsilon^2}$; On the other hand, we also have $W(\varepsilon, \Delta t) = O(\varepsilon^{-3}\Delta t^2)$ if $\Delta t < \varepsilon^2$. In fact, from
 744 $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$ for all $x \in \mathbb{R}$, we get

$$\begin{aligned} W(\varepsilon, \Delta t) &= \frac{1}{1-\varepsilon} \left[\varepsilon \left(1 - \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\Delta t}{\varepsilon^2}\right)^n \right) - \sum_{n=0}^{\infty} \frac{1}{n!} ((1-\varepsilon)^n - 1) \left(-\frac{\Delta t}{\varepsilon^2}\right)^n \right] \\ &= \frac{1}{1-\varepsilon} \left[-\varepsilon \left(\sum_{n=1}^{\infty} \frac{1}{n!} \left(-\frac{\Delta t}{\varepsilon^2}\right)^n \right) + \sum_{n=1}^{\infty} \frac{1}{n!} (1 - (1-\varepsilon)^n) \left(-\frac{\Delta t}{\varepsilon^2}\right)^n \right] \\ &= \sum_{n=2}^{\infty} \frac{(-1)^n}{n!} (1 - (1-\varepsilon)^{n-1}) \left(\frac{\Delta t}{\varepsilon^2}\right)^n \\ &= \frac{1}{2} \varepsilon^{-3} \Delta t^2 - \sum_{k=1}^{\infty} (a_{2k+1} - a_{2k+2}), \end{aligned}$$

745 where

$$a_{2k+1} = \frac{1 - (1-\varepsilon)^{2k}}{(2k+1)!} \left(\frac{\Delta t}{\varepsilon^2}\right)^{2k+1}.$$

746 We will show that $a_{2k+1} > a_{2k+2}$ for all $k \geq 1$. Let us define

$$u(k) = (2k+2) \frac{1 - (1-\varepsilon)^{2k}}{1 - (1-\varepsilon)^{2k+1}}. \quad (\text{B.4})$$

747 We can verify that $u'(k) > 0$ for all $k \geq 1$ if $0 < \varepsilon < 1$, so $u(k)$ is monotonically increasing.
 748 Then $u(k) \geq u(1) = \frac{4(1-(1-\varepsilon)^2)}{1-(1-\varepsilon)^3} > 1 > \Delta t/\varepsilon^2$. This implies that $a_{2k+1} > a_{2k+2}$ for any $k \geq 1$.
 749 Hence $W(\varepsilon, \Delta t) < \frac{1}{2} \varepsilon^{-3} \Delta t^2$ if $\Delta t < \varepsilon^2$.

750 In summary, we arrive at an upper bound for $W(\varepsilon, \Delta t)$,

$$W^*(\varepsilon, \Delta t) = \begin{cases} \frac{\Delta t^2}{2\varepsilon^3}, & \text{if } \Delta t \leq \varepsilon^2; \\ \frac{\varepsilon}{1-\varepsilon}, & \text{if } \Delta t > \varepsilon^2. \end{cases} \quad (\text{B.5})$$

751 As a consequence, we get that for $\varepsilon \in (0, 1)$, $\|\mathbf{F} - \mathbf{G}\|_\infty = O(\varepsilon^{-3}\Delta t^2)$ if $\Delta t \leq \varepsilon^2$ and
 752 $\|\mathbf{F} - \mathbf{G}\|_\infty = O(\frac{\varepsilon}{1-\varepsilon})$ otherwise. Furthermore, from the Figure B.1, it shows that $W^*(\varepsilon, \Delta t)$ is
 753 a very sharp upper bound of $W(\varepsilon, \Delta t)$.
 754 □

755 Appendix C. Proof of Theorem 4.2

756 *Proof.* First, we consider the SL1-LDG q method (3.32) and examine how each term contributes
 757 to the amplification matrix. For the diffusion term, with (4.1), we have

$$\begin{aligned} r_{I_i}(\tilde{\rho}_h^{n+1}, \phi_i^j) &= \sum_{l=0}^{q-1} \tilde{\rho}_{i,l}^{n+1} \phi_i^l(x_{i+\frac{1}{2}}^-) \phi_i^j(x_{i+\frac{1}{2}}^-) - \sum_{l=0}^{q-1} \tilde{\rho}_{i-1,l}^{n+1} \phi_{i-1}^l(x_{i-\frac{1}{2}}^-) \phi_i^j(x_{i-\frac{1}{2}}^+) \\ &\quad - \sum_{l=0}^{q-1} \tilde{\rho}_{i,l}^{n+1} \int_{I_i} \phi_i^l(x) \partial_x \phi_i^j(x) dx. \end{aligned} \quad (\text{C.1})$$

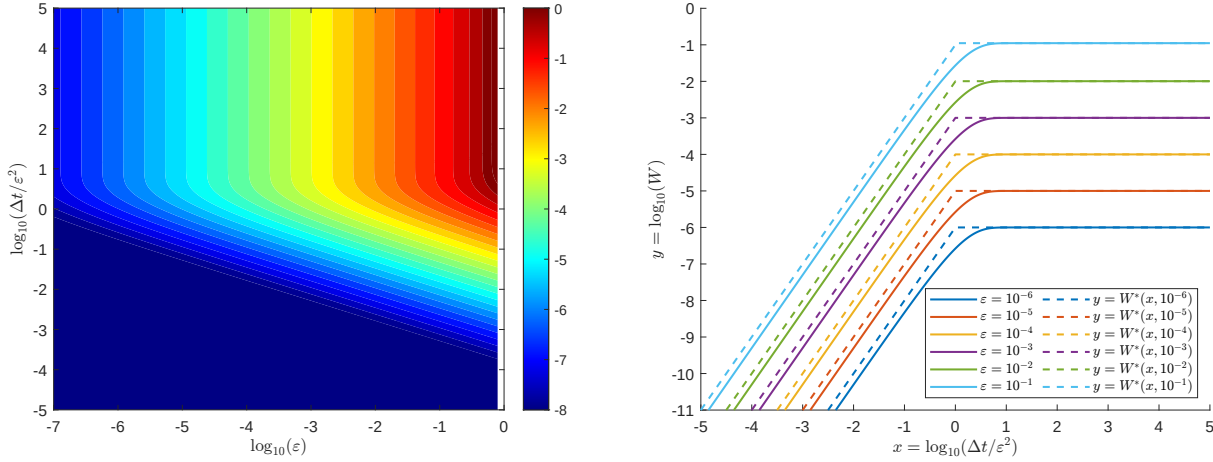


Figure B.1: Left: 2D contour of $W(\varepsilon, \Delta t)$; Right: comparison of $W(\varepsilon, \Delta t)$ and $W^*(\varepsilon, \Delta t)$. A colored solid line represents the slice of W at a given ε , while a dashed line with the same color represents the slice of upper bound W^* at the same ε .

758 Substituting the ansatz $\tilde{\rho}_{i,l}^n = \hat{\rho}_l^n \exp(\mathcal{I}\kappa x_i)$ into (C.1) and use $\phi_i^l(x) = \phi^l(\xi_i)$ with $\xi = \frac{x-x_i}{h}$, we
 759 obtain

$$r_{I_i}(\tilde{\rho}_h^{n+1}, \phi_i^j) = \exp(\mathcal{I}\kappa x_i) \sum_{l=0}^{q-1} R_{j,l}(\omega) \hat{\rho}_l^{n+1}, \quad (\text{C.2})$$

760 where

$$R_{j,l}(\omega) = \phi^l\left(\frac{1}{2}\right)\phi^j\left(\frac{1}{2}\right) - \exp(-\mathcal{I}\omega)\phi^l\left(\frac{1}{2}\right)\phi^j\left(-\frac{1}{2}\right) - \int_{-1/2}^{1/2} \phi^l(\xi)\partial_\xi\phi^j(\xi)d\xi.$$

761 We denote $\mathbf{R}(\omega) = (R_{j,l}(\omega)) \in \mathbb{R}^{q \times q}$. Similarly, there exists $\mathbf{L}(\omega) = (L_{j,l}(\omega)) \in \mathbb{R}^{q \times q}$, such that
 762

$$(\mathcal{D}_h^q q_h^{n+1}, \phi_i^j)_{I_i} = \exp(\mathcal{I}\kappa x_i) \sum_{l=0}^{k-1} L_{j,l}(\omega) \hat{q}_l^{n+1}. \quad (\text{C.3})$$

763 For the convection term, we denote $x^* = x - v\Delta t/\varepsilon$ and define an operator $\mathcal{D}_{h,v}^g : U_h^{q-1} \rightarrow U_h^{q-1}$
 764 for a given positive velocity $v > 0$ (the case of negative velocity is similar), such that

$$\begin{aligned} (\mathcal{D}_{h,v}^g g_h^n(\cdot, v), \phi_i^j)_{I_i} &= \sum_{l=0}^{q-1} g_{i+m_{S-1},l}^n \phi_{i+m_{S-1}}^l(x_{i+\frac{1}{2}}^{*, -}) \phi_i^j(x_{i+\frac{1}{2}}^-) \\ &\quad - \sum_{l=0}^{q-1} g_{i+m_0,l}^n \phi_{i+m_0}^l(x_{i-\frac{1}{2}}^{*, -}) \phi_i^j(x_{i-\frac{1}{2}}^+) \\ &\quad - \sum_{s=0}^{S-1} \sum_{l=0}^{q-1} g_{i+m_s,l}^n \int_{J_{i,s}} \phi_{i+m_s}^l(x^*) \partial_x \phi_i^j(x) dx. \end{aligned} \quad (\text{C.4})$$

765 Here we assume that an element I_i at t^{n+1} is splitted into S segments, denoted by $J_{i,s}$ for
 766 $s = 0, \dots, S-1$. Each $J_{i,s}$ is tracked as an upstream segment $J_{i,s}^*$ contained in an element I_{i+m_s}
 767 at t^n , resulting in an upstream element I_i^* . Thanks to the constant velocity v , I_i is not out of
 768 shape when tracking along characteristic lines, so that $S \leq 2$ in the one-dimensional case. We

769 claim that there exists $\mathbf{H}(\omega, \frac{\Delta t}{\varepsilon h}; v) \in \mathbb{R}^{q \times q}$, such that

$$(\mathcal{D}_{h,v}^g g_h^n(\cdot, v), \phi_i^j)_{I_i} = \exp(\mathcal{I} \kappa x_i) \sum_{l=0}^{q-1} H_{j,l}(\omega, \frac{\Delta t}{\varepsilon h}; v) \widehat{g}_l^n. \quad (\text{C.5})$$

770 (1) When $S = 1$, $\frac{v\Delta t}{\varepsilon h}$ is an integer, then (C.4) can be written as (C.5) with

$$\begin{aligned} H_{j,l}(\omega, \frac{\Delta t}{\varepsilon h}; v) &= \exp(\mathcal{I} m \omega) \phi^l(\frac{1}{2}) \phi^j(\frac{1}{2}) - \exp(\mathcal{I} (m-1) \omega) \phi^l(\frac{1}{2}) \phi^j(-\frac{1}{2}) \\ &\quad - \exp(\mathcal{I} m \omega) \int_{-\frac{1}{2}}^{\frac{1}{2}} \phi^l(\xi) \phi^j(\xi) d\xi, \quad \text{where } m = -\frac{v\Delta t}{\varepsilon h}. \end{aligned} \quad (\text{C.6})$$

771 (2) When $S = 2$, we set $J_{i,0} = [x_{i-\frac{1}{2}}, \hat{x}]$ and $J_{i,1} = [\hat{x}, x_{i+\frac{1}{2}}]$. With a given velocity v , we first
 772 find the reference coordinate $\hat{\xi} = \frac{\hat{x} - x_i}{h}$ of $\hat{x}$ with respect to I_i as follows. Let $x_{i-\frac{1}{2}}^* < \hat{x}^* =$
 773 $x_{i+m+\frac{1}{2}}^* < x_{i+\frac{1}{2}}^*$ and then $m = -\lfloor \frac{v\Delta t}{\varepsilon h} \rfloor - 1$,

$$\hat{\xi} = \frac{\hat{x} - x_i}{h} = \frac{\hat{x}^* - x_i^*}{h} = -\frac{1}{2} + \left(\frac{v\Delta t}{\varepsilon h} - \left\lfloor \frac{v\Delta t}{\varepsilon h} \right\rfloor \right) \in [-\frac{1}{2}, \frac{1}{2}]. \quad (\text{C.7})$$

774 Therefore, in the reference coordinate, an element I_i at t^{n+1} is splitted into $\hat{J}_{i,0} = [-\frac{1}{2}, \hat{\xi}]$ and
 775 $\hat{J}_{i,1} = [\hat{\xi}, \frac{1}{2}]$. In addition, an element I_i^* at t^n is splitted into $J_{i,0}^* = [x_{i-\frac{1}{2}}^*, \hat{x}^*]$ and $J_{i,1}^* = [\hat{x}^*, x_{i+\frac{1}{2}}^*]$,
 776 where

$$x_{i+m_0-\frac{1}{2}} < x_{i-\frac{1}{2}}^* < x_{i+m_0+\frac{1}{2}}, \quad x_{i+m_1-\frac{1}{2}} < x_{i+\frac{1}{2}}^* < x_{i+m_1+\frac{1}{2}}. \quad (\text{C.8})$$

777 Let $\lfloor x \rfloor$ denote the floor of x , we get

$$m_0 = -\left\lfloor \frac{v\Delta t}{\varepsilon h} \right\rfloor - 1, \quad m_1 = -\left\lfloor \frac{v\Delta t}{\varepsilon h} \right\rfloor. \quad (\text{C.9})$$

778 We next compute the reference coordinates of $J_{i,s}^*$ with respect to I_{i+m_s} . Firstly, let $x \in J_{i,0}$,
 779 $x^* \in J_{i,0}^*$ and $\xi = \frac{x - x_i}{h} \in \hat{J}_{i,0}$. With (C.7), we get

$$\frac{x^* - x_{i+m_0}}{h} = \frac{x - x_i}{h} - \left(\frac{v\Delta t}{\varepsilon h} - \left\lfloor \frac{v\Delta t}{\varepsilon h} \right\rfloor - 1 \right) = \xi - \hat{\xi} + \frac{1}{2} \in [-\hat{\xi}, \frac{1}{2}] =: \hat{J}_{i,0}^*. \quad (\text{C.10})$$

780 This means that the reference coordinate of a point $x^* \in J_{i,0}^*$ with respect to I_{i+m_s} locates
 781 within $\hat{J}_{i,0}^*$. Similarly, we can derive that

$$\frac{x^* - x_{i+m_1}}{h} = \frac{x - x_i}{h} - \left(\frac{v\Delta t}{\varepsilon h} - \left\lfloor \frac{v\Delta t}{\varepsilon h} \right\rfloor \right) = \xi - \hat{\xi} - \frac{1}{2} \in [-\frac{1}{2}, -\hat{\xi}] =: \hat{J}_{i,1}^*. \quad (\text{C.11})$$

782 At this end, we can rewrite (C.4) in the form of (C.5) with

$$\begin{aligned} H_{j,l}(\omega, \frac{\Delta t}{\varepsilon h}; v) &= \exp(\mathcal{I} m_1 \omega) \phi^l(-\hat{\xi}) \phi^j(\frac{1}{2}) - \exp(\mathcal{I} m_0 \omega) \phi^l(-\hat{\xi}) \phi^j(-\frac{1}{2}) \\ &\quad - \exp(\mathcal{I} m_0 \omega) \int_{-\frac{1}{2}}^{\hat{\xi}} \phi^l(\xi - \hat{\xi} + \frac{1}{2}) \partial_x \phi^j(\xi) d\xi \\ &\quad - \exp(\mathcal{I} m_1 \omega) \int_{\hat{\xi}}^{\frac{1}{2}} \phi^l(\xi - \hat{\xi} - \frac{1}{2}) \partial_x \phi^j(\xi) d\xi. \end{aligned} \quad (\text{C.12})$$

783 According to (C.5), there exists $\mathbf{S} = \sum_{\alpha=1}^{N_v} \mathbf{B}_\alpha$ and $\mathbf{B}_\alpha(\omega, \frac{\Delta t}{\varepsilon h}; w_\alpha, v_\alpha)$ ($1 \leq \alpha \leq N_v$), such
 784 that

$$b_{I_i}(\langle v \mathbf{S}_v^n[g_h^n] \rangle, \phi_i^j) = \exp(\mathcal{I} \kappa x_i) \sum_{\alpha=1}^{N_v} \sum_{l=0}^{q-1} B_{j,l,\alpha}(\omega, \frac{\Delta t}{\varepsilon h}; w_\alpha, v_\alpha) \hat{g}_l^n. \quad (\text{C.13})$$

785 Similarly, there exists $\mathbf{C}_\alpha(\omega, \frac{\Delta t}{\varepsilon h}; v_\alpha) \in \mathbb{R}^{q \times q}$, such that

$$a_{I_i}(f_h^n(\cdot, v), \phi_i^j) = \exp(\mathcal{I} \kappa x_i) \sum_{l=0}^{k-1} C_{j,l,\alpha}(\omega, \frac{\Delta t}{\varepsilon h}; v_\alpha) \hat{f}_l^n. \quad (\text{C.14})$$

786 Now, we can write the SL1-LDG q method as (4.3) with $\mathbf{A}^{1,q} = (\mathbf{A}_L^{1,q})^{-1} \mathbf{A}_R^{1,q}$, where

$$\mathbf{A}_L^{1,q} = \begin{pmatrix} \mathbf{I} - c_2 \mathbf{L} \mathbf{D} & 0 & 0 & 0 \\ -c_4 \mathbf{I} & \mathbf{I} & 0 & 0 \\ -c_4 \mathbf{I} & 0 & \mathbf{I} & 0 \\ 0 & -\frac{1}{2} \mathbf{I} & -\frac{1}{2} \mathbf{I} & \mathbf{I} \end{pmatrix}, \quad \mathbf{A}_R^{1,q} = \begin{pmatrix} 0 & c_1 \mathbf{B}_1 & c_1 \mathbf{B}_2 & -c_1 \mathbf{S} \\ 0 & c_3 \mathbf{C}_1 & 0 & 0 \\ 0 & 0 & c_3 \mathbf{C}_2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{C.15})$$

787 $\mathbf{I}$ is the identity matrix and

$$\begin{aligned} c_1 &= \frac{\Delta t}{\varepsilon h} \exp\left(-\frac{\Delta t}{\varepsilon^2}\right) = \frac{\Delta t}{\varepsilon h} \exp\left(-\frac{\Delta t/(\varepsilon h)}{\varepsilon/h}\right), \\ c_2 &= \frac{\Delta t}{h^2} \left(1 - \exp\left(-\frac{\Delta t}{\varepsilon^2}\right)\right) = \frac{\Delta t}{\varepsilon h} \left(1 - \exp\left(-\frac{\Delta t/(\varepsilon h)}{\varepsilon/h}\right)\right), \\ c_3 &= \frac{\varepsilon^2}{\varepsilon^2 + \Delta t} = \frac{\varepsilon/h}{\varepsilon/h + \Delta t/(\varepsilon h)}, \\ c_4 &= \frac{\Delta t}{\varepsilon^2 + \Delta t} = \frac{\Delta t/(\varepsilon h)}{\varepsilon/h + \Delta t/(\varepsilon h)}. \end{aligned} \quad (\text{C.16})$$

788 This implies that $\mathbf{A}^{1,q}$ depends only on $\frac{\varepsilon}{h}$ and $\frac{\Delta t}{\varepsilon h}$. For the SL2-LDG q method (3.33), we can
 789 carry out a similar analysis as above. We can obtain $\mathbf{A}^{2,q} = (\mathbf{A}_L^{2,q})^{-1} \mathbf{A}_R^{2,q}$, where

$$\begin{aligned} \mathbf{A}_L^{2,q} &= \begin{pmatrix} 3\mathbf{I} - 2c_2 \mathbf{L} \mathbf{D} & 0 & 0 & 0 & 0 \\ -c_6 \mathbf{I} & \mathbf{I} & 0 & 0 & 0 \\ -c_6 \mathbf{I} & 0 & \mathbf{I} & 0 & 0 \\ 0 & -\frac{1}{2} \mathbf{I} & -\frac{1}{2} \mathbf{I} & \mathbf{I} & 0 \\ 0 & 0 & 0 & 0 & \mathbf{I} \end{pmatrix}, \\ \mathbf{A}_R^{2,q} &= \begin{pmatrix} 0 & 2c_1 \mathbf{B}_1 & 2c_1 \mathbf{B}_2 & 4\mathbf{I} - 2c_1 \mathbf{S} & -\mathbf{I} \\ 0 & (c_5 - c_6) \mathbf{C}_1 & 0 & c_6 \mathbf{C}_1 & 0 \\ 0 & 0 & (c_5 - c_6) \mathbf{C}_2 & c_6 \mathbf{C}_2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{I} & 0 \end{pmatrix}, \\ c_5 &= \frac{2\varepsilon^2}{2\varepsilon^2 + \Delta t} = \frac{2\varepsilon/h}{2\varepsilon/h + \Delta t/(\varepsilon h)}, \\ c_6 &= \frac{\Delta t}{2\varepsilon^2 + \Delta t} = \frac{\Delta t/(\varepsilon h)}{2\varepsilon/h + \Delta t/(\varepsilon h)}. \end{aligned} \quad (\text{C.17})$$

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